Caspian Journal of Applied Mathematics, Ecology and Economics
V. 10, No 2, 2022, December
ISSN 1560-4055

Nodal Solutions of Certain Boundary Value Problem for Fourth Order Ordinary Differential Equations

N.A. Neymatov, R.G. Poladov

Abstract. In this paper, we consider a nonlinear boundary value problem for fourth order ordinary differential equations. We show the existence of two different solutions of this problem with a fixed number of simple zeros.

Key Words and Phrases: nonlinear problem, eigenvalue, eigenfunction, bifurcation point, simple zero, global continua

2010 Mathematics Subject Classifications: 34B05, 34B15, 34C23, 47H11, 47J10, 47J15

1. Introduction

We consider the following nonlinear boundary value problem

$$\ell(y) \equiv (p(x)y'')'' - (q(x)y')' = \varrho\,\tau(x)g(y(x)),\ x \in (0,l), \tag{1.1}$$

$$\begin{aligned} &y'(0)\cos\alpha - (py'')(0)\sin\alpha = 0,\\ &y(0)\cos\beta + Ty(0)\sin\beta = 0,\\ &y'(l)\cos\gamma + (py'')(l)\sin\gamma = 0,\\ &y(l)\cos\delta - Ty(l)\sin\delta = 0, \end{aligned} \tag{1.2}$$

$Ty \equiv (py'')' - qy'$, $p \in C^2\left([0,l];(0,+\infty)\right)$, $q \in C^1\left([0,l];[0,+\infty)\right)$, $\tau \in C\left([0,l];(0,+\infty)\right)$, ϱ is a positive parameter, and α, β, γ, δ are real constants such that $0 \leq \alpha,\ \beta, \gamma,\ \delta \leq \frac{\pi}{2}$ (the cases $\alpha = \gamma = 0$, $\beta = \delta = \pi/2$ and $\alpha = \beta = \gamma = \delta = \pi/2$ are excluded). The nonlinear term $g \in C^0\left([0,l] \times \mathbb{R}^5; \mathbb{R}\right)$ and satisfies the following conditions:

(B1) $sg(s) > 0$ for any $s \in \mathbb{R}$, $s \neq 0$;

(B2) there exist positive constants g_0 and g_∞ such that

$$g_0 = \lim_{|s|\to 0} \frac{g(s)}{s},\ g_\infty = \lim_{|s|\to +\infty} \frac{g(s)}{s}.$$

It is well known that nonlinear boundary value problems for fourth-order ordinary differential equations arise when modeling various problems in mechanics, physics, and various areas of natural science, see [1, 3, 5] and references therein. Note that there are many papers that are devoted to the existence of positive or sign-changing solutions to such problems addressed by using different methods [1-4, 5-16].

Problems similar to the problem (1.1), (1.2) for ordinary differential equations of second order was considered in [6, 7, 10, 11]. In these papers, using the results on the global bifurcation from zero and infinity of nonlinear Sturm-Liouville problems obtained by Rabinowitz [14, 15], Berestycki [4] and Rynne [16], intervals were found such that for each value of the parameter $\varkappa$ from these intervals, the problems under consideration have solutions with fixed usual nodal properties. Note that special cases of the boundary value problem (1.1), (1.2) were considered in [6, 8, 9, 12, 13], where the values of the parameter $\varkappa$ were found, for which the problems have positive and negative solutions [6, 8, 13] or have solutions with a certain number of sign change nodes [9, 12].

In the present paper, we will find an interval for $\varkappa$, in which there are solutions to problem (1.1), (1.2) with a fixed number of oscillations.

The rest of this paper is arranged as follows. In Section 2, we study the behavior of the nonlinear term g on the neighborhoods of zero and infinity using the conditions from B2. In Section 3, using global bifurcation results of [1, 2], we determine the interval of $\varkappa$, in which there exist nodal solutions of nonlinear boundary value problem (1.1), (1.2).

2. Preliminary and some properties of function g

Let BC be the set of functions that satisfy the boundary conditions (1.2), and let $E = C^3([0,1];\mathbb{R}) \cap BC$ be the Banach space with the norm

$$||y||_3 = \sum_{s=0}^{3} ||y^{(s)}||_\infty, \; ||y||_\infty = \max_{x\in[\,0,l]} |y(x)|.$$

By the first part of (B2) we have the following representation

$$g(s) = g_0 + \tilde{g}_0(s), \tag{2.1}$$

where

$$\lim_{|s|\to 0} \frac{\tilde{g}_0(s)}{s} = 0. \tag{2.2}$$

Lemma 2.1. *The following relation holds:*

$$||\tilde{g}_0(y)||_\infty = o(||y||_3) \text{ as } ||y||_3 \to 0. \tag{2.3}$$

Proof. Let

$$\chi(y) = \max_{0\le|s|\le y} |\tilde{g}_0(s)|.$$

Then $\chi(y)$ is a nondecreasing function on $(0, +\infty)$. Moreover, by (2.2) we have the relation

$$\lim_{y\to 0+} \frac{\chi(y)}{y} = 0. \tag{2.4}$$

Since χ is nondecreasing we get the following inequalities

$$\frac{\tilde{g}_0(y)}{||y||_3} \leq \frac{\chi(|y|)}{||y||_3} \leq \frac{\chi(||y||_\infty)}{||y||_3} \leq \frac{\chi(||y||_3)}{||y||_3},$$

whence, with regards (2.4), implies that

$$\frac{||\tilde{g}_0(y)||_\infty}{||y||_3} \to 0 \text{ as } ||y||_3 \to 0.$$

The proof of this lemma is complete.

By the second part of (B2) we have the following representation

$$g(s) = g_\infty + \tilde{g}_\infty(s), \tag{2.5}$$

where

$$\lim_{|s| \to +\infty} \frac{\tilde{g}_\infty(s)}{s} = 0. \tag{2.6}$$

Lemma 2.2. *One has t following relation:*

$$||\tilde{g}_\infty(y)||_\infty = o(||y||_3) \text{ as } ||y||_3 \to \infty. \tag{2.7}$$

Proof. We define the function $\sigma(y)$, $y \in (0, +\infty)$, as follows:

$$\sigma(y) = \max_{0 \leq |s| \leq y} |\tilde{g}_\infty(s)|.$$

Then $\sigma(y)$ is a nondecreasing function on $(0, +\infty)$. Moreover, it follows from (2.6) that

$$\lim_{y \to +\infty} \frac{\sigma(y)}{y} = 0. \tag{2.8}$$

Since χ is nondecreasing the following relations hold:

$$\frac{\tilde{g}_\infty(y)}{||y||_3} \leq \frac{\sigma(|y|)}{||y||_3} \leq \frac{\sigma(||y||_\infty)}{||y||_3} \leq \frac{\chi(||y||_3)}{||y||_3},$$

Hence, due to (2.8), we obtain

$$\frac{||\tilde{g}_0(y)||_\infty}{||y||_3} \to 0 \text{ as } ||y||_3 \to 0.$$

The proof of this lemma is complete.

Lemma 2.3. *There exist a positive constants c_0 and c_∞ such that*

$$c_0 \leq \frac{g(s)}{s} \leq c_\infty \text{ for } s \in \mathbb{R},\ s \neq 0. \tag{2.9}$$

Proof. Let $\epsilon_0 > 0$ be a sufficiently small fixed number such that

$$g_0 - \epsilon_0 > 0 \text{ and } g_\infty - \epsilon_0 > 0.$$

Then, by (B2), there exist a sufficiently small number $b_0 > 0$ and a sufficiently large number $d_\infty > 0$ such that

$$g_0 - \epsilon_0 < \frac{g(s)}{s} < g_0 + \epsilon_0 \text{ for } |s| < d_0,\ s \neq 0 \tag{2.10}$$

and

$$g_\infty - \epsilon_0 < \frac{g(s)}{s} < g_\infty + \epsilon_0 \text{ for } |s| > d_\infty. \tag{2.11}$$

Moreover, since $\frac{f(s)}{s}$ is continuous on $[-d_\infty, -d_0] \cup [d_0, d_\infty]$, there are positive constants κ_0 and κ_∞ such that

$$\kappa_0 \leq \frac{f(s)}{s} \leq \kappa_\infty \text{ for } d_0 \leq |s| \leq d_\infty. \tag{2.12}$$

Let

$$c_0 = \min\{g_0 - \epsilon_0,\ g_\infty - \epsilon_0,\ \kappa_0\} \text{ and } c_\infty = \min\{g_0 + \epsilon_0,\ g_\infty + \epsilon_0,\ \kappa_\infty\}.$$

Then (2.9) follows directly from (2.10)-(2.12). The proof of this lemma is complete.

3. Main results

We consider the following linear eigenvalue problem

$$\begin{cases} \ell(y)(x) = \lambda\tau(x)y(x),\ x \in (0,l), \\ y \in BC. \end{cases} \tag{3.1}$$

where $\lambda \in \mathbb{C}$ is a spectral parameter. By [3, Theorems 5.4, 5.5] the eigenvalues of problem (3.1) are positive, simple and form an infinitely increasing sequence $\{\lambda_k\}_{k=1}^{\infty}$. Moreover, for each $k \in \mathbb{N}$ the eigenfunction y_k corresponding to the eigenvalue λ_k has exactly $k-1$ simple zeros.

To study the global bifurcation of solutions of nonlinear perturbations of problem (3.1), in [1] the author constructed classes S_k^ν, $k \in \mathbb{N}$, $\nu \in \{+, -\}$, of functions in E that have the oscillation properties of eigenfunctions (and their derivatives) of the linear problem (3.1). Note that the sets S_k^+, S_k^- and $S_k = S_k^+ \cap S_k^-$ are pairwise disjoint open subsets of E, and if $y \in \partial S_k^\nu$ (∂S_k), then by [1, Lemma 3.1] y has at least one zero of multiplicity 4 in $(0,l)$.

Alongside the boundary-value problem (1.1), (1.2) we shall consider the following nonlinear eigenvalue problem

$$\begin{cases} \ell(y)(x) = \lambda\varrho\,\tau(x)g(y(x)),\ x \in (0,l), \\ y \in BC. \end{cases} \tag{3.2}$$

It is obvious that any solution of (3.2) of the form $(1,y)$ yields a solution y of (1.1), (1.2). Below we will show that for some $k \in \mathbb{N}$ and every $\nu \in \{+, -\}$ there exists a solution to problem (1.1), (1.2) of the form $(1,y)$ with $y \in S_k^\nu$.

Theorem 3.1. *For each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ there exists a continuum $C_{k,0}^{\nu}$ of solutions of problem (1.1), (1.2), containing $\left(\frac{\lambda_k}{\varrho g_0}, 0\right)$ is unbounded in $\mathbb{R} \times E$ and lies in $(\mathbb{R} \times S_k^{\nu}) \cup \left\{\left(\frac{\lambda_k}{\varrho g_0}, 0\right)\right\}$.*

Proof. By (2.1) problem (3.2) takes the following form:

$$\begin{cases} \ell(y)(x) = \lambda \varrho\, g_0\, \tau(x) y(x) + \lambda \varrho\, \tau(x) \tilde{g}_0(y(x)), \ x \in (0, l), \\ y \in BC. \end{cases} \tag{3.3}$$

It follows from (2.1) that

$$||\lambda \varrho\, \tau\, \tilde{g}_0(y)||_{\infty} = o(||y||_3) \text{ as } ||y||_3 \to 0, \tag{3.4}$$

uniformly in $\lambda \in \Lambda$ for any bounded interval $\Lambda \subset \mathbb{R}$. Hence problem (3.3) is linearizable and the corresponding linear problem

$$\begin{cases} \ell(y)(x) = \lambda \varrho\, g_0\, \tau(x) y(x), \ x \in (0, l), \\ y \in BC. \end{cases} \tag{3.5}$$

possesses infinitely many eigenvalues $\tilde{\lambda}_1 < \tilde{\lambda}_2 < \ldots < \tilde{\lambda}_k \to +\infty$, all of which are simple. The eigenfunction $\tilde{y}_k$ corresponding to $\tilde{\lambda}_k$ lies in S_k. Moreover, from (3.5) it can be seen that $\tilde{\lambda}_k = \frac{\lambda_k}{\varrho g_0}$. Then it follows from [1, Theorem 1.1] that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ there exists a continuum $C_{k,0}^{\nu}$ of solutions of problem (3.3) which contains $(\tilde{\lambda}_k, 0)$, lies in $(\mathbb{R} \times S_k^{\nu}) \cup \{(\tilde{\lambda}_k, 0)\}$ and is unbounded in $\mathbb{R} \times E$. The proof of this theorem is complete.

By (2.5) problem (3.2) can be rewritten in the following form:

$$\begin{cases} \ell(y)(x) = \lambda \varrho\, g_{\infty}\, \tau(x) y(x) + \lambda \varrho\, \tau(x) \tilde{g}_{\infty}(y(x)), \ x \in (0, l), \\ y \in BC. \end{cases} \tag{3.6}$$

Then, using (2.7) and following the appropriate reasoning in the proof of Theorem 2.1, from [2, Theorem 3.1] we obtain the following result.

Theorem 3.2. *For each $k \in \mathbb{N}$ and each ν there exist a continuum $C_{k,\infty}^{\nu}$ of solutions of problem (3.6) (or (3.2)) which contains $\left(\frac{\lambda_k}{\varrho g_{\infty}}, \infty\right)$ and has the following properties:*

(i) *there exists a neighborhood Q_k of $\left(\frac{\lambda_k}{\varrho g_{\infty}}, \infty\right)$ in $\mathbb{R} \times E$ such that*

$$Q_k \cap \left(C_{k,\infty}^{\nu} \backslash \left\{\left(\frac{\lambda_k}{\varrho g_{\infty}}, \infty\right)\right\}\right) \subset \mathbb{R} \times S_k^{\nu};$$

(ii) *either $C_{k,\infty}^{\nu}$ meets $\left(\frac{\lambda'_k}{\varrho g_{\infty}}, \infty\right)$ through $\mathbb{R} \times S_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$, or $C_{k,\infty}^{\nu}$ meets $(\lambda, 0)$ for some $\lambda \in \mathbb{R}$, or projection of $C_{k,\infty}^{\nu}$ on $\mathbb{R} \times \{0\}$ is unbounded.*

Now we can prove the following very important result.

Theorem 3.3. *For each $k \in \mathbb{N}$ and each ν one has the relation:*

$$C_{k,0}^{\nu} = C_{k,\infty}^{\nu}. \tag{3.7}$$

Proof. Since g satisfies both conditions B2, by Lemma 3.1 of [1] it follows from [1, Lemma 1.1] that

$$C^{\nu}_{k,\infty}\backslash\left\{\left(\frac{\lambda_k}{\varrho g_\infty},\infty\right)\right\}\subset\mathbb{R}\times S^{\nu}_k \tag{3.8}$$

(see also [16, Theorem 3.3]). Consequently, the first part of assertion (ii) of Theorem 3.2 cannot hold. Moreover, it follows from [16, Theorem 3.3] that if $C^{\nu}_{k,\infty}$ meets $\mathbb{R}\times\{0\}$, for some $\lambda\in\mathbb{R}$, then $\lambda=\frac{\lambda_k}{\varrho g_0}$. Similarly, if $C^{\nu}_{k,0}$ meets $\mathbb{R}\times\{\infty\}$, for some $\lambda\in\mathbb{R}$, then $\lambda=\frac{\lambda_k}{\varrho g_\infty}$.

If the third part of assertion (ii) of this theorem holds, then there exists a sequence $\{(\lambda^*_n,y^*_n)\}^{\infty}_{n=1}\subset C^{\nu}_{k,\infty}\backslash Q_k$ such that

$$\lambda^*_n\to\pm\infty \text{ as } n\to\infty. \tag{3.9}$$

Since $(\lambda^*_n,y^*_n)\in\mathbb{R}\times S^{\nu}_k$ it follows from (3.2) that λ^*_n is kth eigenvalue of the linear problem

$$\begin{cases}\ell(y)(x)=\lambda\varrho\,\tau(x)h_n y(x),\ x\in(0,l),\\ y\in BC,\end{cases} \tag{3.10}$$

where

$$h_n(x)=\begin{cases}\frac{g(y^*_n(x))}{y^*_n(x)} & \text{if } y^*_n(x)\neq 0,\\ g_0 & \text{if } y^*_n(x)=0.\end{cases} \tag{3.11}$$

On the base (3.11) from (2.9) we get

$$0<c_0\le h_n(x)\le c_\infty,\ x\in[0,l]. \tag{3.12}$$

By (3.12) it follows from [1, relations (4.3) and (4.4)] that the eigenvalues of problems (3.10) are bounded from below uniformly with respect to $n\in\mathbb{N}$, and, consequently, the relation $\lambda^*_n\to-\infty$ is impossible as $n\to\infty$. If $\lambda^*_n\to+\infty$, then for any sufficiently large $n\in\mathbb{N}$ by (3.12) Theorems 5.4 and 5.5 of [3] implies that the number of zeros of the function $y^*_n(x)$ will be sufficiently large, which contradicts the condition $y^*_n\in S^{\nu}_k$. Therefore, the third part of assertion (ii) of Theorem 3.2 does not hold.

Thus, the second part of assertion (ii) of Theorem 3.2 holds. Consequently, $C^{\nu}_{k,\infty}$ meets $(\frac{\lambda_k}{\varrho g_0},0)$ and $C^{\nu}_{k,0}$ meets $(\frac{\lambda_k}{\varrho g_\infty},\infty)$, which, by (3.8), implies that $C^+_{k,0}=C^+_{k,\infty}$ and $C^-_{k,0}=C^-_{k,\infty}$ for any $k\in\mathbb{N}$. The proof of this theorem is complete.

For simplicity, we introduce the notation:

$$C^+_k=C^+_{k,0}=C^+_{k,\infty},\ C^-_k=C^-_{k,0}=C^-_{k,\infty},\ k\in\mathbb{N}.$$

By Theorem 3.1 and 3.3, for each $k\in\mathbb{N}$ and each $\nu\in\{+,-\}$ the continuum C^{ν}_k lies in $(\mathbb{R}\times S^{\nu}_k)\cup\left\{\left(\frac{\lambda_k}{\varrho g_0},0\right)\right\}\cup\left\{\left(\frac{\lambda_k}{\varrho g_\infty},\infty\right)\right\}$, and meets $\left(\frac{\lambda_k}{\varrho g_0},0\right)$ and $\left(\frac{\lambda_k}{\varrho g_\infty},\infty\right)$ in $\mathbb{R}\times E$. It follows from here that if

$$\frac{\lambda_k}{\varrho g_0}<1<\frac{\lambda_k}{\varrho g_\infty} \text{ or } \frac{\lambda_k}{\varrho g_\infty}<1<\frac{\lambda_k}{\varrho g_0} \tag{3.13}$$

for some $k \in \mathbb{N}$, then the continua C_k^+ and C_k^- intersect the hyperplane $\{1\} \times E$ in $\mathbb{R} \times E$. Consequently, for this k there exist solutions $\hat{y}_k^+$ and $\hat{y}_k^-$ of problem (1.1), (1.2) such that $\hat{y}_k^+ \in S_k^+$ and $\hat{y}_k^- \in S_k^-$.

It is obvious that the conditions from (3.13) are equivalent to the following conditions

$$\frac{\lambda_k}{g_0} < \varrho < \frac{\lambda_k}{g_\infty} \text{ or } \frac{\lambda_k}{g_\infty} < \varrho < \frac{\lambda_k}{g_0}, \tag{3.14}$$

respectively. Therefore, we have proved the following main theorem of this paper.

Theorem 3.4. *Let for some $k \in \mathbb{N}$ condition (3.14) holds. Then there exist solutions $\hat{y}_k^+$ and $\hat{y}_k^-$ of problem (1.1), (1.2) such that $\hat{y}_k^+ \in S_k^+$ and $\hat{y}_k^- \in S_k^-$.*

References

[1] Z.S. Aliyev, Global bifurcation of solutions of certain nonlinear eigenvalue problems for ordinary differential equations of fourth order, *Sb. Math.* **207**(12) (2016), 1625-1649.

[2] Z.S. Aliyev, N.A. Mustafayeva, Bifurcation of solutions from infinity for certain nonlinear eigenvalue problems of fourth-order ordinary differential equations, *Electron. J. Differ. Equ.* **2018** (2018), No. 98, 1-19.

[3] D.O. Banks, G. J. Kurowski, A Prüfer transformation for the equation of a vibrating beam subject to axial forces, *J. Differential Equations* **24**(1) (1977), 57-74.

[4] H. Berestycki, On some nonlinear Sturm-Liouville problems, *J. Differential Equations* **26**(3) (1977), 375-390.

[5] B.B. Bolotin, *Vibrations in technique: Handbook in 6 volumes, The vibrations of linear systems. I*, Engineering Industry, Moscow (1978) [in Russian].

[6] G. Dai, X.Han, Global bifurcation and nodal solutions for fourth-order problems with sign-changing weight, *Appl. Math. Comp.* **219** (2013), 9399-9407.

[7] G. Dai, R. Ma, Bifurcation from intervals for Sturm-Liouville problems and its applications, *Electron. J. Differ. Equ.* 2014(3) (2014), 1-10.

[8] J.R.L. Webb, G. Infante, D. Franco, Positive solutions of nonlinear fourth-order boundary-value problems with local and non-local boundary conditions, *Proc. Roy. Soc. Edinburgh Sect. A* **138**(2) (2008), 427-446.

[9] A.C. Lazer, P.J. McKenna, Global bifurcation and a theorem of Tarantello, *J. Math. Anal. Appl.* **181**(3) (1994), 648-655.

[10] R. Ma and G. Dai, Global bifurcation and nodal solutions for a Sturm-Liouville problem with a nonsmooth nonlinearity, *J. Funct. Anal.* **265** (2013), 1443-1459.

[11] R. Ma, B. Thompson, A note on bifurcation from an interval, *Nonlinear Anal.* **62**(4) (2005), 743-749.

[12] R. Ma, B. Thompson, Nodal solutions for a nonlinear fourth-order eigenvalue problem, *Acta Math. Sin. (Engl. Ser.)* **24**(1) (2008), 27-34

[13] R. Ma, Jia Xu, Bifurcation from interval and positive solutions of a nonlinear fourth-order boundary value problem, *Nonlinear Anal.* **72**(1) (2010), 113-122.

[14] P.H. Rabinowitz, Some global results for nonlinear eigenvalue problems, *J. Funct. Anal.* **7**(3) (1971), 487-513.

[15] P. H. Rabinowitz, On bifurcation from infinity, *J. Differential Equations* **14**(3) (1973), 462-475.

[16] B.P. Rynne, Bifurcation from zero or infinity in Sturm-Liouville problems which are not linearizable, *J. Math. Anal. Appl.* **228**(1) (1998), 141-156.

Nazim A. Neymatov
Ganja State University, Ganja, Azerbaijan
E-mail: nneymatov@yandex.ru

Rovshan G. Poladov
Baku State University, Baku, Azerbaijan
E-mail: r_poladov@mail.ru

Received 27 August 2022
Accepted 29 September 2022

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 10, No 2, 2022, December
ISSN 1560-4055

An Approach to Analysis and Classification of Medical Errors

G.G.Abdullayeva, H.R.Nagızade

Abstract. The paper is devoted to organization of paperless technologies in in medicine, especially in the first and urgent medical care in the Republic. Application of information technologies increased the importance of electronic information exchange, which in turn it enables to develop paperless work process. Today it can be argued that electronic information prevails in many fields social action and its priority is the development of paperless technology. The protocol forms suggected in the paper imply continuous information transfer, connecting them with the center and connection of the station that is very important with polyclinics, city hospitals, information exchange. The system provides error detection to ensure patent's safety, the study and examing a doctor's errors and construction of fair information exchange in hospitals and outpatient practices.

Key Words and Phrases: automation, doctor's error, mathematical statistics, frequency analysis, protocol

2010 Mathematics Subject Classifications: 34B05, 34B15, 34C23, 47H11, 47J10, 47J15

1. Introduction

A civilized society naturally means qualitative, that is, efficient strategic planning and based on personnel, medical, technical and information policy, a society to provide state guarantee for the provision of necessary medical services [1]. Application of information technologies increased importance of electronic information exchange, which in turn allows to develop a paperless work process. Today we can argue that electronic information prevails in many fields of social action and its priority is the development of paperless technology.

Increasing role of information and communication technologies (ICT) could not bypass this area of health care. Its application in healthcare is on the one hand can significantly remove fiscal constraints in the provision of medical services. On the other , digital ICTs can enable consultations and facilitate daily tests and information transfer for patients and most importantly, can save clinicians from a large amounts of paperwork.

Mistakes of medical personnel has a particularly negative effect on the ability of a person to realize their human rights, freedoms and legal interests. The contradictions about the quality of medical services show that the people in need of medical care often become hostages of the prevailing stereotype connected with inevitability of medical errors. Specialists who harm the patients, in order to avoid spiritual and legal responsibilities, show their mistakes as one of the natural directions in the development of the existing professional status.

The argument in favor of the inadmissibility of the side effects of the activities of medical workers is mainly based on the fact that existing legislation, science, medicine and law enforcement practice, with the content characteristics of the medical services category an uniquivocal position has not yet been formed.

Objective assessment of the level and dynamics of crimes because of negligence is complicated by the lack of official statistics on the damage to the patient interests. The observed trend creates a significant social threat. A growing sense of irresponsibility among incompetent medical workers for mistakes made cannot ensure the safety of human life and his health. This, in turn causes an increase in the cases of neglecting the diagnostic and treatment standards.

Even in the most developed, that is, industrialized countries, health care is not safe as it should be. This issue remains a problem. True, the 1970-s and 1980-s there was a sharp increase in medical malpractice lawsuits started to happen. This was especially felt in the United States [1]. So, the Harvard Medical Practice Study [2,3], the researches conducted in Colorado and Utah [4] systematically show where the patients were often harmed. Collected data on hospitals and the results of both studies in the United States formed the basis of the report on the quality of health care. In 2000, in the article "To err is human" [5] Klaus Deter Scheppokat and Johann Neu note that the number of people who died in a year as a result of a doctor's error in America was between 44000-98000. Subsequently, many developed countries conducted similar research [6] and created relevant institutions. The goal to improve the patient level of security. Of these in Great Britain National Parient Safety Agency (NPSA), in Germany German Coalition for Patient Safety (Aktionsbündnis Patientensicherheit, APS) organizations can be named.

According to the All-Russian Public Opinion Research Center, today 60% of patients in Russia are dissatisfied with the quality of medical care. It is quite right that the public price is low for the level of medical care. Compulsory Medical Insurance Fund reports that about 10% of medical care in Russia is defective.

Acute social nature of the medical malpractice problem is complicated by the fact that law enforcement agencies and the court pay little attention to the harmful activities of incompetent health workers [7]. Only in recent years some steps have been taken to eliminate this problem. So, in order to improve patient safety in the first step, three basic strategies were proposed:

1. reporting errors by the security control system,
2. studying and investigating errors
3. setting fair information exchange in hospitals and doctor's outpatient practices.

If these are performed critical events can be identified, reported and analyzed so that similar incidents can be prevented and measures taken. Every happened event must be necessarily evaluated. Finally, if any preventable adverse events occur, appropriate measures to prevent damage should be taken.

The public is also very interested in this problem. For example, based on the 2005 year's survey, 72% of citizens of the European Union, 78% of German and European Union are dissatisfied with medical errors and in general they consider the existence of these types of errors to be an important problem. 28% of them are patients that expressed theirconcern after the treatment they received [8].

2. Problem statement

The article discusses the problem of organizing paperless technology in first urgent medicine for complete medical care especially throughout the republic.

Development of such a system considers uninterrupted data transmission to networks of substations, their center coordination and as a very important issue, communication of the station with hospitals and clinics, and information exchange. And the goal is to determine the shortcomings of the existing electronic system in the emergency medical service based on the study of the whole period, to revise the work process, to take into account requirements of society as a whole and to develop a new electronic solution throught evalution.

3. Problem solving

Successful completion of the problem requires solving a number of sub-tasks. Among these sub-tasks the main ones are:

Conducting proper preliminary analytics for formation and analysis of process and requirements;

Preparation of a functional document based on the created requirements (technical tasks);

Technical development of the software base;

Design of compatible screens;

Preparation of the internal and external part of the registration department (for the internal part 4 screens, and 4 screens for the outer part);

The final completion of the first phase of the software and the preparing a temporary login screen;

Early stage of software testing and recording the results of the software;

Systematic analysis of medical errors;

Presentation and first testing of the software in the Emergency Medical Station and production in the test environment, etc.

More than 40 forms of tables for various purposes have been developed [9]. Let's mention some of them:

Filters related to the large doctor service window:

Start date	End date	Auxiliary aid station	Doctor	Individual identification number	Patient's name	Patient's last name	
Call adres	District	Settlement	Type of transportation	Type of call	Diagnosis name	Hospital	See the result

Figure 1.

1. (a) organization of the medical team (disruption of work functions: did not come to work, extension called, working time is over);
   (b) functions of doctors:

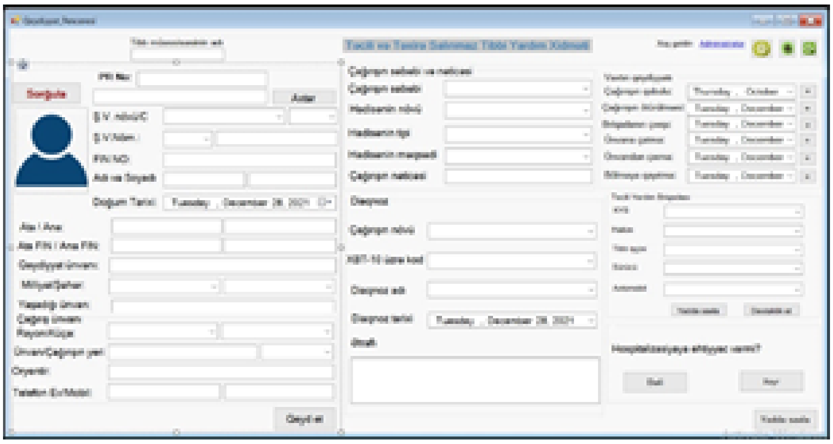


Figure 2. Fragment from the doctor's tablet

1. (a) types of accident; survere cases; call points, etc.
   (b) the result of the call:

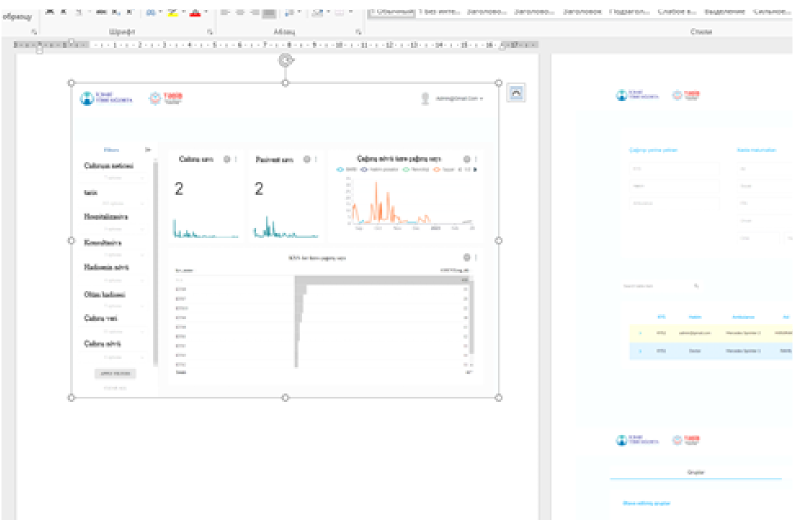


Figure 3. Analysis of results

In general, since 1990 doctros in developed countries, national and international discussions devoted to the problem of minimizing and at best eliminating errors are being conducted. Forward as the main context the safety of the patient is put. Errors encountered are classified, classes are expanded and discussions are held to prevent adverse events, quality standards are being developed, and the topic of raising the level of health care services is the basis of meetings..

Based on todays accepted context, doctro's mistakes can be divided into the groups indicated in table 1.

Table 1. Overview of physician errors

?	Event	Explanation
1	Adverse event	More treatment-induced adverse events-may ot may not be preventable
2	A preventable adverse event	The negative event that can be prevented
3	Critical event	An event resulting in a negative event or an event with an increased probability of occurrence
4	Error	Action or inaction that leades to deviation fron the goal, following a wrong plan, lack of planning
5	Partial ignore	A mistake that does not cause much damage

A medical malpractice, whether large or small, both quantitative and qualitative should be examined and evaluated from a qualitative point of view. Evalution of its performance is equivalent to evaluting the functional

$$A = A\{N;\ N_i;\ N_t\ \ (i \geq t); D_i^m;\ Y_i^m;\ Q_i^m\} \qquad (1)$$

Here A is the doctor of call team;

N is the number of working days of the doctor in a month;

N_i the number of real calls of the doctor during one working day;

$N_t\ \ (i \geq t)$ the number of times the doctor attended the call directly without intervention (false calls, refusal of the doctor, absence of the patient at the call address, etc);

D_i^m initial diagnosis made by the doctor;

Y_i^m provided first aid;

Q_i^m the decision making by the doctor.

But we cannot forget that the problems in the doctor's work are inevitable, especially in the work of emergency physicians. For example, the information about the calls of the Emergency and first Medical Aid Station in Baku city in 2020-2021 is given in table 2. Based on the data, it can be seen that what kinds of calls doctors face every day.

Table 2. Events encountered by the medical team

?	**Call events**	**2020**		**2021**	
		During the year	**1 day**	**During the year**	**1 day**
1	Unsuccessful call	63555	175	62424	171
2	False calls	1015	3	1206	4
3	Not found at location	6206	17	6310	17
4	Address not found	749	2	1076	3
5	Death before the doctor	16044	44	16852	46
6	Unreasonable call	39541	108	36980	101

The effectiveness of the organization of medical assistance to the population mainly depends on how adequately the influences of the regional features are taken into account. Therefore, it is necessary to analyze the activity of ambulatory and emergency medical services at the regional level, and then to develop measures aimed at improving their work. We believe that in application evidence-based measures and methods the role of modern information technologies is undeniable.

At different stages when evaluating the quality of medical care to patients we need to stop on a concept such as inconsistency in diagnoses. According to discrepancy between diagnoses, etiology of the disease and the nature of the pathological process (for example, according to the nature of stroke) we can meet two cases. So, low diagnosis (presence of another nosology) and extreme diagnosis (absence of this nosology) can be classified. Differences in the diagnoses may also be related to late diagnosis.

Within these terms it is appropriate to conduct a selective analysis of the ratio of similar diagnoses in the provision of medical care at different stages. In different years (2010-2020), although the share of discrepancies in diagnoses by selection method was observed with the decrease trend, it varied between 0.9-4.1%. At the same time, it is necessary to survey among doctors to investigate the reason for the existence of these cases, to define the main reasons for the discrepancy between the diagnoses. It is important to determine the issue, because at the same time several reasons (combination of objective and subjective reasons) the result containing non informative and makes further analysis extremely difficult.

In diagnoses in the assessment of diseases by the doctors the main reasons for non-compliance are the short duration of the patient stay in the medical institution, the difficulty in diagnosing the disease, as well as hiding manifestation of a number of nosological forms.

According to the questionnaire survey of doctor's survey table was developed for the objective analysis of the objective reasons discrepancies in diagnoses (percentage of positive answers).

Outpatient, polyclinic, emergency and urgent care organizations doctors are asked to answer to the following types of questions: they are collected; lack of information to assess the patient's state, in some cases incompleteness of diagnostic equipment or apparatus, hidden form of the disease, the rarity or severity of the disease.

It should be noted that the cases of inconsistency in the diagnoses of the emergency and first aid station and the hospital can be explained by a number of reasons – time shortage, extreme situation, absense of additional laboratory, diagnostic methods of research, mainly the lack of paramedic staff of mobile groups, and unexpected situation. When analyzing inconsistency in diagnoses based on a questionnaire survey of doctors the subjective reasons especially inifficient examination of the patient, incorrect assessment of anamnestic data, clinical underreporting or incompletereporting of information should not be overlooked.

Since the work is devoted to the development of paperless technologies it is intended to create intellectual information system through which it implements uninterrupted transmission of information to networks of substations, their connecting with the center and very important, theconnection of the station with polyclinic, city clinics and exchange information with hospitals and clinics. As we mentioned above, different purpose forms have been developed, doctors have been provided with tablets. Each medical team includes to this table the tima spends on call, the initial diagnosis, the actions and decision taken. Thus, a set of data allows monitoring the work of each medical team.

For example, the work of the medical teams of the divisions 6, 1, 20, 4, 8, 2, 13, 16, 22, 3, 5, 7, 9, 11, 18, 19 of Baku Emergency during a month was analyzed, that is functional (**??**) was evaluted. The information filled in for each call on the doctors tablet is examined.

44 defective cases were detected. 2% of them - anesthesiological reaminatology brigde without reguest (ARB) was called; 4.5% - the situation was not evaluted correctly and resulted in death at the doctor; 6.8% - diagnosis does not it the complaint; 18% - incomplete medical care; 15.9% - acute cerebral circulation disorder, to reduce arterial pressblood pressure is not targeted, drug extravagance, assistance tactics were not chosen correctly, etc.; 13% - the diagnosis was not made correctly; 26.7% - symptoms confirming the diagnosis were not fully checked; 13.1% - incompletely filled out accompanying sheet (protocol form on tablet).

Created system for detection and elimination of medical errors can be considered as an approach. The frequency of errors, along with ensuring their classification can be submitted to the commission that makes decisions objectively for the doctors in the form of advices (taking into account legislative reguirements).

The system can always submit such type of analysis to the decision making person.

4. Main result

The development of paperless technology in healthcare is an unseparable part of telemedicine. From this point of view, this system can be considered as a basis for remote transmission of information followed by all ethical moral norms and legal rights and at the same time for its safe strage. When the system detects doctors error, in one hand he can sort (systematize) and analyze them and on the other hand, to give consultative advice.

References

[1] Cooper J.K., Egeberg R.O., Stephens S.K., Where is the malpractice crisis taking us. West J. Med. 1977;127: 262–6.

[2] Brennan T.A., Leape L.L, Laird N.M, et al.: Incidence of adverse events and negligence in hospitalized patients: results of the Harvard Medical Practice Study I. N Eng. J. Med 1991; 324: 370–6.

[3] Leape L.L., Brennan T.A., Laird N., et al.: The nature of adverse events in hospitalized patients. Results of the Harvard Medical Practice Study II. N Eng. J. Med 1991; 324: 377–84.

[4] Gawande A.A., Thomas E.J., Zinner M.J., Brennan T.A., The incidence and nature of surgical adverse events in Colorado and Utah in 1992. Surgery 1999; 126: 66–75.

[5] Klaus Dieter Scheppokat, Johann Neu. Medical Data and Quality Management, Dtsch Arztebl 2007; 104(46): A 3172–7 www.aerzteblatt.de

[6] Kohn L.T., Corrigan J.M., Donaldson M.S. (eds.); Committee on Quality in Health Care; Institute of Medicine: To err is human. Building a safer health system. Washington: National Academy Press, 1999.

[7] Kudakov A.V., Medical error and its legal-criminal assessment. Dissertation and thesis according to Higher Attestation Commission of RF 12.00.08, PhDs in Law.

[8] Barbara Hoffmann, Julia Rohe. Patient Safety and Error Management //Deutsches Ärzteblatt International | Dtsch Arztebl Int 2010; 107(6): 92–9

[9] Abdullayeva G.G., Nagizade H.R., Role of information technologies in organizing functioning of emergency and first aid, The XI international scientific practical Conference "Symmetries: theoretical and metodical aspects", Aztrakhan, september, 2022, p.7-14.

G.G.Abdullayeva
Institute of Control Systems of the Ministry of Science and Education
E-mail: ag_gulchin@rambler.ru

H.R.Nagızade
Institute of Mathematics and Mechanics of the Ministry of Science and Education
E-mail: hasan.nagizada@yahoo.com

Received 25 August 2022
Accepted 01 October 2022

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 10, No 2, 2022, December
ISSN 1560-4055

Poisoning with Carbon Monoxide and Importance of Monitoring of Patients after CO Poisoning

Irada H. Mirzazadeh

Abstract. Carbon monoxide is known to be a highly dangerous indoor pollutant leading to severe health outcomes. However, CO-related mortality data are not available through standard reporting schemes, and therefore, the magnitude of CO-related mortality has always been subject to estimation. Along with the diagnosis of poisoning by carbon monoxide poisoning, forecast has a great importance for the consequences of monitoring. It is to necessary in observe during a certain time health status of persons poisoned, as carbon monoxide poisoning can cause damage to tissues and organs such as the cardiovascular and respiratory systems, muscles, liver, and kidneys. The study identifies the problems that exist with the suggests actions for better monitoring.

Key Words and Phrases: carbon monoxide, monitoring, nervous and cardiovascular systems

2010 Mathematics Subject Classifications: Primary 68N19, 68P10, 68T20

---

1. Introduction

Indoor air pollution sources from outdoor diffusion and indoor human activities have become the leading cause of the disease burden. Numerous poisons, from natural toxins to synthetic chemicals existing in our environment, can produce a wide variety of deleterious effects in living organisms. Usually, indoor air has the same pollutants as outdoor and could cause the same disease effects, such as particulate matter (PM), sulphur dioxide (SO2), nitrogen dioxide (NO2), and carbon monoxide (CO). However, the health effect and the emergency poisoning event caused by ambient CO were rare, while this usually occurred in an indoor environment with limited space and high concentration has become an urgent public health problem worldwide. This review article discusses carbon monoxide poisoning from a clinical point of view. CO sometimes termed a "silent killer" is a colourless, odorless, and non-irritable gas. As the specific gravity of CO is 0.97, it is slightly lighter than air. CO normally present in the atmosphere at a concentration of approximately 0.03–0.20 parts per million. This gas can be generated by natural or anthropogenic sources, especially by incomplete combustion of fossil fuels and biomasses. The toxicity of CO is mostly attributable to its capability to diffuse into the erythrocytes and bind to haemoglobin, for which CO has a nearly 200-fold higher affinity than oxygen.

In poisoned patients, both oxygen and CO compete for binding to haemoglobin, but even small volumes of CO dissolved in the blood may be effective to largely displace oxygen binding (i.e. the so-called Haldane effect) [1]. The severity of symptoms is conventionally determined by the exposure values of CO and the consequent carboxyhaemoglobin (COHb) level, whose normal concentration in blood is typically comprised between 0.5% and 1.5% in non-smokers [2]. A CO concentration between 10% and 20% is usually accompanied by nausea, fatigue, tachypnoea, emotionality, confusion, and clumsiness, CO levels between 21% and 30% is accompanied with headache, exertional dyspnoea, angina, visual impairment, and decreased sensory perception, and that between 31% and 50% is accompanied with dizziness, fainting, confusion, nausea, vomiting, visual impairment, and problematic decision-making, while CO values >51% typically trigger seizures, coma, severe acidosis, and death [3]. Toxic exposures more frequently occur from faulty heaters, fires, industrial accidents, and car exhausts.

2. Worldwide epidemiology of carbon monoxide poisoning

This article presents updated information on the worldwide burden of carbon monoxide (CO) poisoning. The worldwide epidemiologic data were obtained from the Global Health Data Exchange registry, a large database of health-related data maintained by the Institute for Health Metrics and Evaluation. The worldwide cumulative incidence and mortality of CO poisoning are currently estimated at 137 cases and 4.6 deaths per million, respectively. The worldwide incidence has remained stable during the last 25 years, while both mortality and percentage of patients who died have declined by 36% and 40%, respectively. The incidence of CO poisoning does not differ between sexes, whilst mortality is double in men. The incidence shows two apparent peaks, between 0–14 years and 20–39 years. The percentage of patients who died constantly increases in parallel with aging, peaking in patients aged 80 years or older. The number of CO poisoning grows in parallel with the socio-demographic index (SDI), though more detailed analyses would be needed to confirm our findings. Mortality displays a similar trend, being approximately 2.1- and 3.6-fold higher in middle and middle-to-high than in low-to-middle SDI countries [4].

Carbon monoxide (CO) poisoning is a public health concern in developing countries especially in China with a high disease burden. A total of 27 855 non-occupational CO poisoning cases were reported between 2017 and 2021 in China [5]. In the recent 5 years, 90% of the CO poisoning cases of China occurred in the northern region, and Shandong Province got the first with nearly half cases reported [5, 6]. Jinan city with a 10 million population is the capital of Shandong province. In Jinan between 2007 and 2021 among total of 6588 CO poisoning, 5616 cases (85.25%) and 180 deaths caused by household coal heating was identified during study period. The cumulative incidence rate was 5.78 per 100,000 person-years and the mortality rate was 0.19 per 100,000 person-years. The incidence in urban areas (6.55 per 100,000 person-years) was higher than rural areas (5.04 per 100,000 person-years), and there was a statistical difference between urban and rural ($P < 0.001$) ($P < 0.001$). The poisoning time point mainly occurs in the sleep stage [7].

Since the mid-1950s, Korean society started to use coal briquettes as fuel for cooking and heating. According to a survey of the leading causes of accidental CO poisoning between 1965 and 1976, gas leaking from ondol structures accounted for 54.2% of accidents, while gas leaking from a fireplace such as a coal fuel hole accounted for 26.3%. Overall, about 80% of CO poisoning cases could be attributed to ondol heating facilities [8]. CO poisoning incidence increased during the 1960s. This trend had continued until the 1980s. The number of victims of CO poisoning peaked at approximately 1 million people in 1980. The number of mild cases in 1980 at approximately 0.86 million cases, while severe cases numbered approximately 134,000 that year. The number of fatal cases first exceeded 1,000 in 1964, and it was estimated that at least 1,000 annual deaths continued during the subsequent 30 years, until 1993. The number of fatal cases peaked in 1980, with approximately 2,400 deaths [9].

In the 1990s, CO poisoning was dramatically reduced as coal briquettes were replaced by oil. However, in Korea CO poisoning using burning coal briquettes during suicide attempts has recently increased. The number of suicides associated with CO poisoning were 34 among the total 10,653 suicide cases (0.3%) in 2006, but 1125 was observed among the total 14,159 suicidal cases (7.9%) in 2012 [10]. We estimated that, from 1951 to 2018, the cumulative number of CO poisoning victims comprised approximately 22,830,000 mild cases, 3,570,000 severe cases, and 65,000 deaths.

Carbon monoxide (CO) is the leading cause of poisoning deaths in many countries, including Japan. Annually, CO poisoning claims about 2000-5000 lives in Japan (Fig. 1), which is over half of the total number of poisoning deaths. However, deaths from CO poisoning doubled in 2003 compared to that for 2001 in Japan. This may be attributable to an increase of suicides by means of CO inhalation [11]. We estimated that, in Japan, 42339 persons died from the toxic effects of CO between 1997 and 2009. Eighty-three percent of the deaths occurred in males [12].

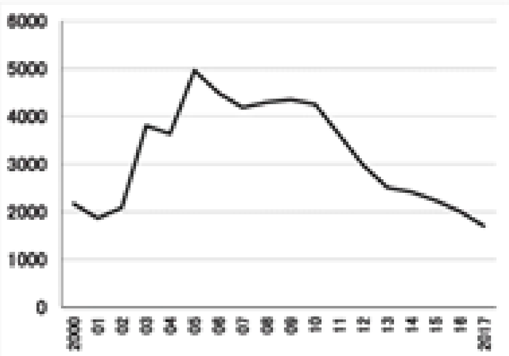


Fig. 1. Statistics on annual numbers of death by CO poisoning in Japan (2000–2015).

In England, unintentional non-fire related poisoning (UNFR) is the most common cause of CO poisoning, with an average of 25 deaths per year being reported between 2015 and 2016 by the Office of National Statistics (ONS) (Fig. 2) [13, 14] and a hospitalization rate of 0.49/100,000 between 2001 and 2010 [15]. Between 2002 and 2016 were identified 6643 hospital admissions for CO poisoning in England, excluding fire-related cases (n = 408; 5.9%) (Fig. 3). Of these, 1782 (52.5%) were unintentional, which represented 47.5% (n =

1617) and 52.5% (n = 1782) of female and male hospitalizations, respectively (p = 0.091). Overall, this is equivalent on average to 227 unintentional non-fire related CO poisoning hospital admissions per year (min. 2013 = 166; max. 2010 = 326). There was clear seasonality in the admissions, with the largest proportion of hospitalisations occurring during winter months (November to February) [16].

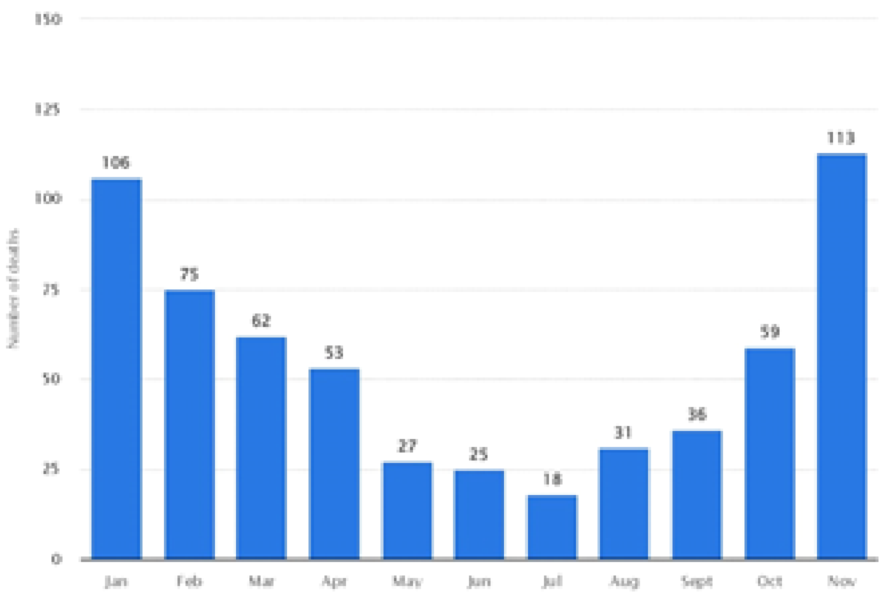


Fig. 2. Number of deaths from unintentional carbon monoxide poisoning in the United Kingdom (UK) from 1995 to 2021, by month

The highest age-specific crude rates were found in poisoning with carbon monoxide among those aged over 85 years. Crude rates among children aged < 10 years were slightly higher than those found in older children, aged from 10 to 24 and young adult groups, aged from 25 to 29. Here were found higher age-standardized rates in rural areas compared to urban areas. These patterns were observed both overall and by gender. Age-standardized rates were slightly higher among males than females, respectively [16].

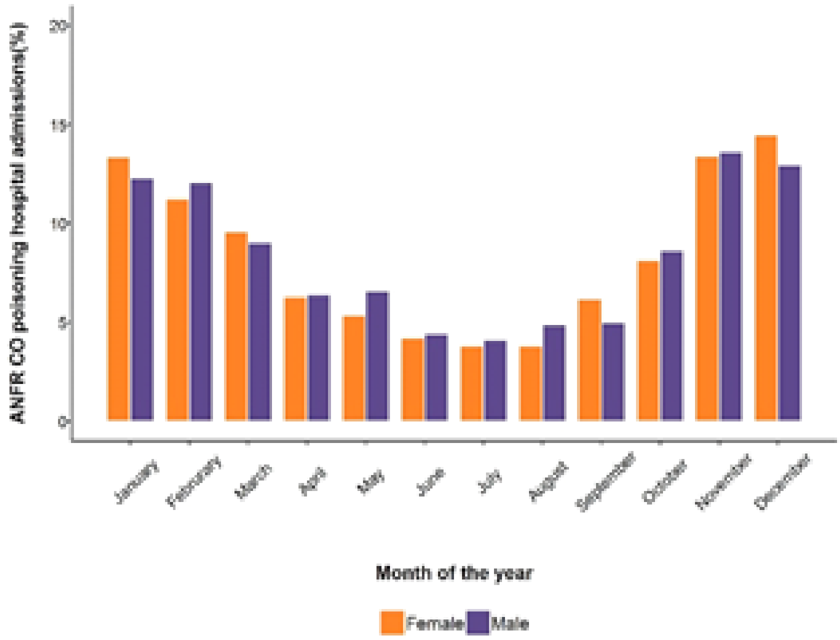


Fig. 3. Percentage of ANFR CO poisoning hospital admissions among males (purple) and females (orange) by calendar month, England, 2002–2016.

In England, temporal trends were analysed using piecewise log-linear models and comparing them to analogous data obtained for Canada, France, Spain, and the US. Here were estimated age-standardized rates per 100,000 inhabitants by area-level characteristics using the WHO standard population (2000–2025). Temporal trends showed significant decreases after 2010. Decreasing trends were also observed across all countries studied, yet France had a 5-fold higher risk. Based on 3399 unintentional non-fire related CO poisoning hospitalizations, is found an increased risk in areas classified as rural, highly deprived or with the largest proportion of Asian or Black population [16].

According to Statistics Canada, between 2000 and 2013, in Canada, there were 4,990 deaths associated to CO poisoning. This number included 1,125 deaths where there were no other underlying causes of death and 3,865 where there were other underlying causes of death. In terms of the total number of deaths, 34.8% were registered for people between the ages of 25 and 44, and an additional 40% were reported for those between the ages of 45 and 64. Quebec had the highest total number of carbon monoxide-related deaths (n = 1,445) followed by Ontario (27.5 per cent), the Prairies (24.6 per cent), British Columbia and the Territories (13.3 per cent), and the Maritimes (5.6 per cent). In total, there were 3,027 hospitalizations related to CO poisoning in Canada between 2002 and 2016, and 14.3% of CO-related hospitalizations in Canada were for people of 65 years of age or older. When missing data or unspecified data were removed from the analysis, 75% of CO-related hospitalizations occurred as a result of CO poisoning originating at home. Ontario, Alberta, and British Columbia had the highest number of CO-related hospitalizations, while the largest increases in per capita hospitalization between 2002 and 2016 were in Saskatchewan (34.7 per cent), Manitoba (19.2 per cent), and British Columbia (7.6 per cent). There are more than 300 CO-related deaths per year in Canada, and more than 200 hospitalizations per year in Canada [17].

In the United States, 50,000 patients with carbon monoxide poisoning are admitted to the emergency departments of hospitals annually, resulting in 1,500 deaths [18, 19]. Most of poisoning events are observed and reported in the state of Nebraska in January [20, 21]. In the period between 2000 and 2009, the number of carbon monoxide poisoning patients estimated was about 68,316; 30,798 (45.1%) were provided on-site aid, and 36,691 (53.7%) patients were treated in hospitals. 34,386 of poisoned patients are women, 30,257 are men. Most of poisoning events are related to the living conditions, and here, a special place have women, children less than 17 years of age, and persons aged between 18 and 44. In spite of these factors the quantity of poisoned persons has decreased: in 2006 - 0.31%, in 2009 - 0.24%. Between 2000 and 2009, 16,447 death events are registered [22, 23]. In 2015, according to data from the Centers for Disease Control and Prevention (CDC) in the United States, there were 393 deaths from accidental carbon monoxide poisoning (Fig. 4) [24].

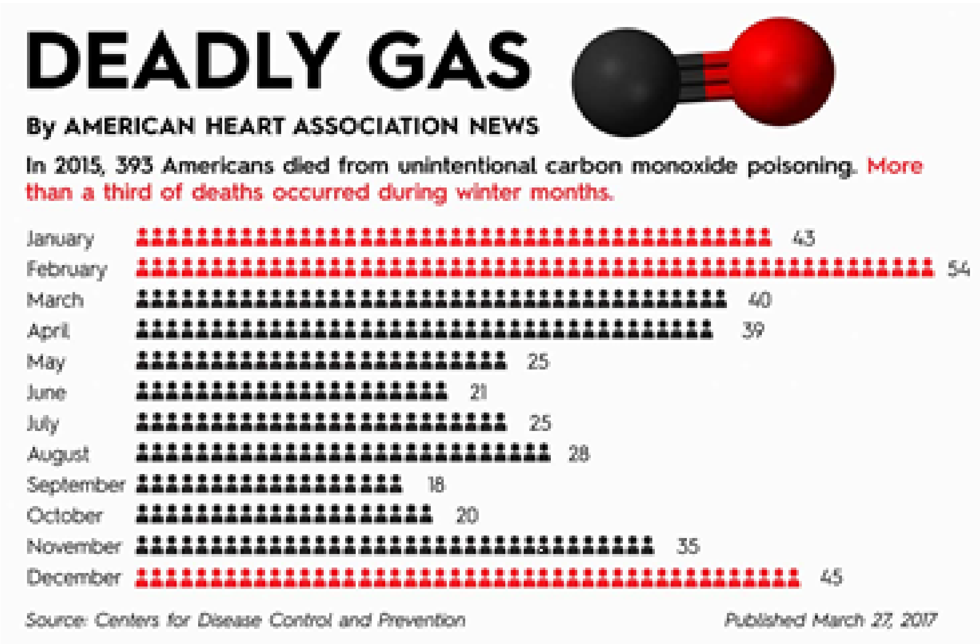


Fig. 4. Statistical information on carbon monoxide poisoning in USA

Between January 2008 to December 2017, the risk of CO-related death in Turkey was 0.35/100000. There were 2667 deaths from CO poisoning in the 10-year period. 1371 (51.4%) of the victims were male, 1178 (44.2%) of them were female and there were 118 (4.4%) victims whose genders were unknown. The median age of the patients was 45 years (range, 15 days-108 years). Most of the deaths occurred in cases of $\geq$50 years of age. 2545 (95.4%) of the incidents were heating related, 50 (1.9%) of them were work related and 72 (2.7%) of them were for unknown cause. There was a stagnating trend of CO-related deaths. Most of the incidents occurred in winter. The Middle Anatolian region was of the highest risk in CO-related mortality [25]. Figure 5. shows mapping of carbon monoxide related death risk in Turkey.

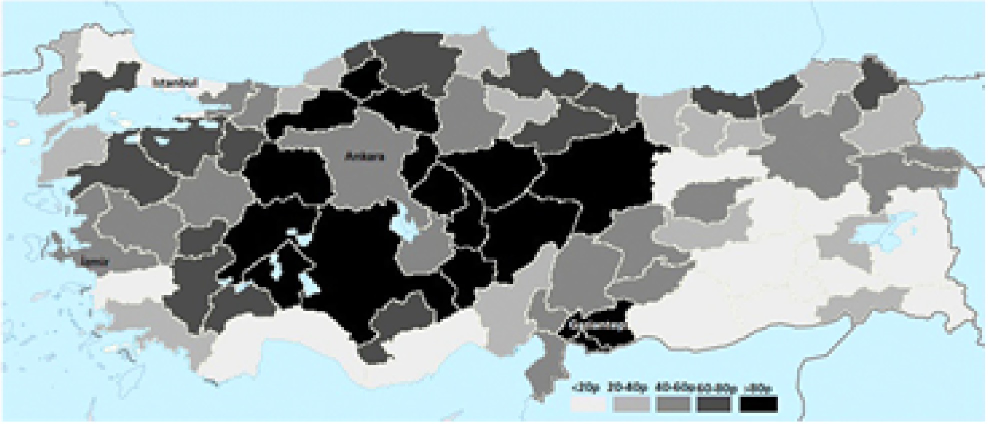


Fig. 5. CO-related death risks in the provinces of Turkey

According to statistical data, a considerable increase in the number of acute poisonings with carbon monoxide has been observed in Azerbaijan too. Between 2010 and 2022, in Baku (the capital of Azerbaijan), the number of carbon monoxide poisoning patients estimated is 21,404. Reliable Information on poisonings within Baku during the period between 2010 and 2022 is given in Table 1.

Table 1.

Statistical information on carbon monoxide poisoning in the districts of Baku

Baku city, districts	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Narimanov	69	121	127	118	105	137	77	42	56	46	41	32	43
Khatai	106	135	192	126	161	201	177	139	138	152	98	104	104
Sabayil	42	85	109	88	139	74	94	84	57	75	65	59	57
Yasamal	118	137	151	132	76	116	183	195	202	188	162	130	173
Nasimi	129	221	237	178	187	185	100	71	86	95	75	68	71
Nizami	64	123	171	200	197	175	159	100	112	154	112	80	129
Binegedi	190	316	395	378	371	419	493	352	345	393	322	277	293
Khazar	17	26	36	63	78	91	106	94	89	147	157	148	142
Surakhani	70	141	141	193	185	185	199	156	162	186	137	198	132
Sabunchi	107	147	221	202	158	177	271	189	241	229	201	238	196
Garadag	98	115	232	145	206	166	139	110	114	105	79	78	77
Total	1010	1567	2012	1823	1863	1949	1998	1532	1602	1770	1449	1412	1417

Most of poisoning events are observed in Baku in winter. Rates of poisoned with CO were higher among females. The cases of poisoned with carbon monoxide among children aged < 10 years old were slightly lower than those found in older children, 10 to 24 years old and young adult groups, 25 to 29 years old. Epidemiology of CO poisoning in different age groups during the period between 2020 and 2022 in Baku is shown in Table 2.

Table 2.

Age-standardized rates of the number of cases of poisoned with CO in Baku (2020–2022) by population characteristics

		2020		**2021**		**2022**	
Age groups (years old)							
	Total	Male	Female	Male	Female	Male	Female
<10	147	18	27	27	30	20	25
10 to 24	1190	144	285	124	213	178	246
25 to 39	1463	210	313	204	335	161	240
40 to 54	751	117	120	117	110	120	167
55 to 69	672	90	105	113	125	112	127
70 to 84	53	3	17	1	13	2	17
>85	2	-	-	-	-	-	2

3. The importance of monitoring by carbon monoxide poisoning

Along with the diagnosis of poisoning by carbon monoxide poisoning in order to forecast has a great importance for the consequences of monitoring. It is to be observed during a certain time health status of persons poisoned. Because we know CO binds to haemoglobin with a higher affinity than oxygen. By interfering with oxygen binding, it induces tissue hypoxia, which is thought to affect the organs that heavily depend on oxygen utilization, including the cardiovascular system [26]. Major causes of tissue and organ damage is not only hypoxia, but also oxidative stress, formation of oxygen reactive species, neuron necrosis, apoptosis, and abnormal inflammation.

The heart is extremely susceptible to CO-induced hypoxia, due to its high oxygen demand. Cardiac involvement manifests mainly as ischemic insult, with elevated enzyme levels and ECG changes ranging from ST-segment depression to transmural infarction. Conduction abnormalities, atrial fibrillation, prolonged QT interval [27] and ventricular arrhythmia have been demonstrated. There have been few studies of CO-induced cardiopulmonary compromise in children. Known is a case of severe cardiopulmonary compromise without overt neuropsychiatric sequelae in a 15-year-old boy [28].

Satran et al. performed a research with 230 patients from 1994 to 2002. They reported that myocardial injury, defined as the elevation of cardiac enzyme levels or ischemic changes indicated in electrocardiography, was observed in 37% of patients with CO poisoning requiring hyperbaric oxygen therapy. When this patient group was followed up until 2005, the hazard ratio (HR) of long-term mortality was significantly higher in the group with myocardial infarction (MI) at 2.1 (95% confidence interval [CI], 1.2–3.7) compared to the group without MI [29]. Additionally, there have been reports of MI after CO poisoning [30–32], and among them, there were cases that occurred at low exposure levels [32]. Therefore, the possibility that the risk of ischemic heart disease (IHD) may increase due to CO poisoning can be considered. However, only a few case reports have confirmed the effects of CO toxicity on the cardiovascular system. Moreover, in a large epidemiological study performed in Taiwan based on a nationwide health database, no increased risk of IHD was found [33]. No large-scale epidemiological studies have explored the relationship between CO poisoning and IHD development other than the study in Taiwan. Therefore, further epidemiological research is needed.

Therefore to investigate the association between CO poisoning and IHD, a nested case-control study of 28,113 patients who experienced CO poisoning and 28,113 controls matched by sex and age was performed using the nationwide health database of South Korea. Based on a conditional logistic regression, there was a significantly higher risk of IHD among the CO poisoning group than among the control group (adjusted hazard ratio [HR], 2.16; 95% confidence interval [CI], 1.87–2.49). IHD risk was the highest within 2 years after CO poisoning. The risk of IHD after CO poisoning was higher among the younger age group under 40 years (adjusted HR, 4.85; 95% CI, 3.20–7.35), and it was much greater among those with comorbidities (adjusted HR, 10.69; 95% CI, 2.41–47.51). The risk of IHD was the highest within the first two years after CO poisoning (adjusted HR, 11.12; 95% CI, 4.54–27.22). Even if more than six years had passed, the risk was still significantly higher than among the control group (adjusted HR, 1.55; 95% CI, 1.27–1.89). The analyses imply that CO poisoning is associated with an increased risk of IHD. Although they observed increased IHD risk after CO poisoning, the exact mechanism of this association remains unclear. Further studies of the long-term effects of CO poisoning on the cardiovascular system and the underlying mechanisms are thus required [34].

There is limited information regarding cardiac imaging after CO poisoning. One case report involving the use of cardiac magnetic resonance imaging (CMRI) detected late gadolinium enhancement (LGE) during the 4-month follow-up of a patient with severe CO poisoning [35]. Figure 6 shows a representative CMRI image of injury patterns and changes. Another report described a patient with acute subendocardial injury after CO poisoning [36].

The symptoms of poisoning with carbon monoxide resolves in most patients following normobaric or hyperbaric oxygen therapy, but a minority of patients experience persistent neuropsychiatric abnormalities or delayed encephalopathy. The carboxyhaemoglobin (COHb) level on admission after CO inhalation does not always correlate with clinical findings or prognosis. Chronic CO exposure may present itself with loss of dentition, gradual-onset neuropsychiatric symptoms, or recent impairment of cognitive ability inconsistent with delayed neurological sequelae (DNS) [37-39]. Increasing evidence indicates that the brain damage caused by CO intoxication results from mitochondrial oxidative stress in the central nervous system, white matter demyelination resulting from the immune response, abnormal inflammatory responses, and apoptosis. Free radicals activate a cascade responsible for the appearance of the delayed effects seen in an estimated 1%-47% of patients with CO intoxication [38, 40].

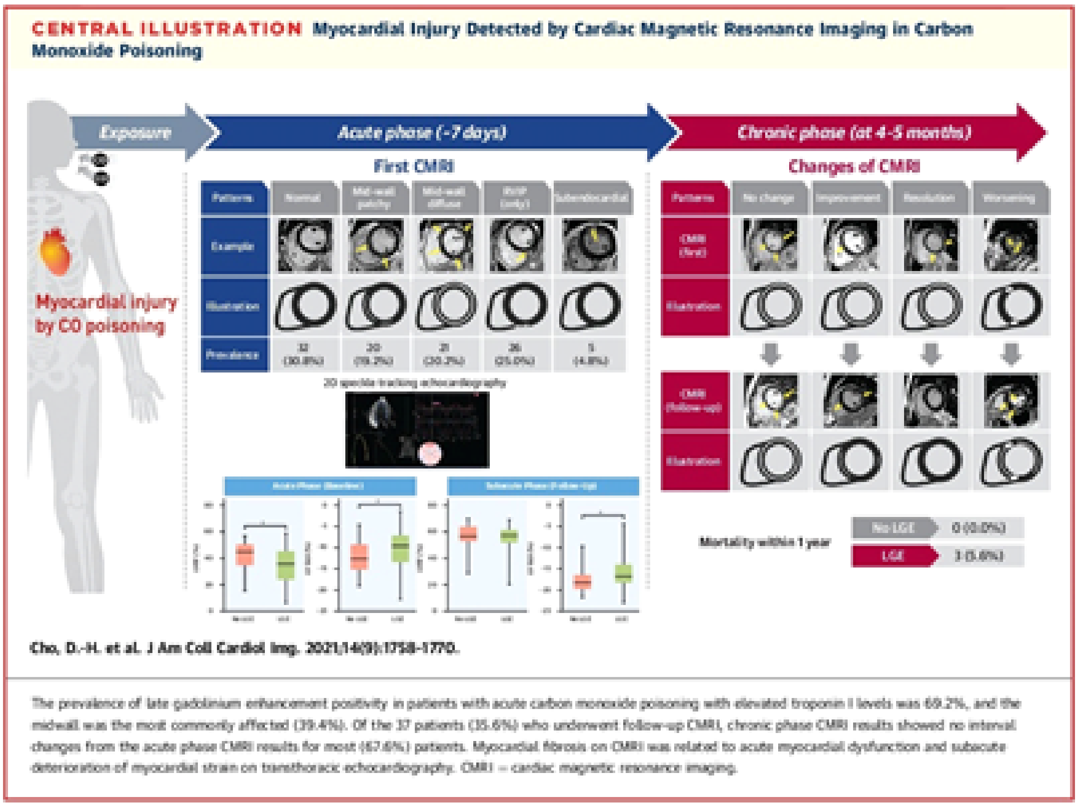


Fig. 6. Representative CMRI image of injury patterns and changes.

DNS includes a broad spectrum of symptoms. The sequelae may vary from mild to severe headache, seizures, alteration in consciousness, lethargy, concentration problems, cognitive disturbances, emotional liability, personality changes, amnestic syndromes, dementia, psychosis, gait disturbances, movement disorders (e.g., parkinsonism), chorea, apraxia, agnosia, inaction, peripheral neuropathy, urinary incontinence, and even vegetative state [38, 40-42]. Delayed CO intoxication is diagnosed by the presence of a clinically silent period or lucid interval lasting for 2-40 days after acute intoxication followed by recurrent neuropsychiatric symptoms [40]. It should also be noted neurological symptoms of CO poisoning can manifest not only immediately but also as late as 2 to 6 weeks.

The diagnosis of DNS is primarily made on the basis of the clinical features and radiological findings of CT and conventional MRI. MRI have shown that particularly vulnerable areas of the brain include the cerebral cortex, hippocampus, basal ganglia, and cerebellum [43]. These include newly formed hyperintense white matter lesions on T2-weighted images and fluid-attenuated inversion recovery (FLAIR), but they may be absent even in cases with chronic neurological symptoms [44, 45]. Cerebrospinal fluid (CSF) analysis is a valuable diagnostic tool, but it is highly invasive and time-consuming.

4. Organization of monitoring, investigation of the dynamics of changes in indicators

Eventually monitoring is advisable for both once poisoned also for persons affected by chronic intoxication.

Monitoring needs to be conducted after successful treatment in hospital. Therefore, starting time of monitoring should coincide with the end of the treatment. Functional parameters and biochemical analysis of carbon monoxide victim needs to be examined from time to time during the monitoring (fixed time interval). Particularly, type of poisoning, more affected poisoning of the body and more nervous and cardiovascular systems, mainly due to the majority of these indicators are checked by selecting among the more specific ones. In most cases, the determination of patient treatment verification of indicators reflecting the health of people is selected in the process of stationar treatment. Analysis in the selected interval of time should be checked for prevention and prognose of consequences after poisoning.

Time changes in carbon monoxide poisoning should be controlled after receiving treatment in order to avoid the consequences of toxication. Time series method is used in those situations. The basis of time series analysis is that former happenings have important indications for future happenings. Time series data is a sequence of successive moments of time, which reflects to the situation. In contrast to randomly selected analysis, time series based on observation data of equal times. Time series can be often found in medicine. Time series analysis has two goals: determination the nature of queue and prognosis. In both cases, the model must be specified before the turn to the interpretation of the data.

According to the analysis of time series, data consists of systematic component and a random voice complication detection components arranged in a regular variable. Majority of research methods allow to observe the change in the index on a regular basis using a variety of methods for filtering noise. Routine variables of time series have two classes: either the trend or seasonal components.

Change dynamics reflects the trend. Trend consists of the variable components changed through the time organized in a systematic linear or non-linear. The seasonal component is repeated periodically [46].

Time series process used to identify prognostic factors of data in the past, today linked to a similar effect in the near future.Analysis of observations is a continuous process estimated in a certain discrete moments of time (when you can evenly across the distribution). For that reason indications which can cause a dangerous development in near future should be selected (months, sometimes years).

There is not "automate" methods for detection of time series. If the trend (increasing or decreasing) is monotone, the queue is not difficult to analyze. If the time series has enough offense in that case smoothing process should be conducted primarily as a method of filtration. Smoothing process is a kind of moderation of data. In this case, the non-systematic errors repel each other. The most common method of smoothing method is moving average, when m members of the neighboring row of each member shall be replaced by a simple average, m is a price of intervals. Also, the trend is to be used for the detection of exponential smoothing. Many monotone time series is described by linear to express analytically. If there is non-linear component, set of data needs to be carried out to remove it. For this reason, most of the time logarithmic, exponential, or polynomial transformations can be used. In some cases, the least squares method is carried out in the smoothing. All of these methods are given the relatively smooth line noise filtering, transforms to circle.

Moving average method determine the start of a new trend, also warns of the end or return. This method allows you to keep track of the development process, it can be viewed as trend lines. However, this method is not used for making predictions, because it follows a trend, but it can't predict, it only shows the start of a new trend. Smoothed curve and the trend observed during the performance of the simplified average, short-term floating-average rate reflect dynamics more accurately for long intervals calculation.

Moving average is defined as follows:

$$y_t = 1/m \tag{1}$$

where y_i, is the value of the i-th level; m is the number of levels from smooth intervals -$(m = 2p + 1)$; y_t dynamic row of the current level; i is the number smooth level range; p is m single range value $p = (m - 1)/2$

Smooth change interval depends on the determination of the indicators. Thus, indicators of irregular, small changes smooth interval assumed to be more. If you are required to take into account changes in smoothing, small gap becomes smaller.

Moving average method is used if time series is organized in straight lines. Because this time does not misrepresent the dynamics of the index. If the range is non-linear, usage of this method can cause distortion of indicators . It is used when smooth is exponential [47].

Analytical smoothing method is an identification of development trends as time series function.

$$\widehat{y_t} = f(t), \tag{2}$$

where $\widehat{y_t}$ is theoretical value of time series with analytical expression for the time, t is time.

Theoretical values are derived from the mathematical model.

Indicating the trend of development, the following features are implemented:

- The linear function with straight line graphs:

$$\widehat{y_t} = a_0 + a_1 t,$$

- Exponential function

$$\widehat{y_t} = a_0 * a_1',$$

- Exponential function second degree (parabola)

$$\widehat{y}_t = a_0 + a_1 * t + a_2 t^2,$$

- Logarithmic function:

$$\widehat{y}_t = a_0 + a_1 lnt$$

Estimation of functions parameters are carried out by least squares method. In this case, the solution is the minimum value of the sum of theoretical and empirical level squares:

$$\sum (\widehat{y}_t - y_i)^2 \to min, \tag{3}$$

where $\widehat{y}$ is calculated, y_t are real levels.

Smooth on a straight line is used in cases where the increments are fixed.

Smooth with exponential function is applied in geometric changes when there is a steady increase in the ratios.

Secondary exponential function smooth is used to change dynamic range and stable chain increases.

The smooth on logarithmic function reflects growth of the number of decrease, the recent increase in the time series.

Counting accuracy of the analytical expressions is defined as follows: sum of empirical series of price must coincide with the sum of the smoothed series levels. In this case, small errors can occur due to the calculated values:

$$\sum y = \sum \widehat{y}_t \tag{4}$$

Autocorrelation is used to determine patterns of additional data change in time series smooth method. Autocorrelation function, determine indication whether it is increasing or decreasing based on seasonal fluctuations.

Determination model is used to assess the trend model accuracy:

$$R^2 = \frac{\sigma^2_{\widehat{y}}}{\sigma^2_y}, \tag{5}$$

where, $\sigma^2_{\widehat{y}}$ is theoretical model dispersion of the data variance, σ^2_y is empirical dispersion of the data.

Trend model shows development tendency of R^2 close to 1 indicators in values.

According to the time series method, data processing is carried out in three stages:

In the first phase filtering is carried out not taking into consideration distortions resulting from seasonal or other changes. The main goal of filtration is to find out y-changes affected from x-changes, eliminate factors that will affect that relationship further. A few known methods for filtering floating above the average value is the most widely used.

According to the moving average price at the time of moving to and from in the price index is calculated by determining the average number. In this situation, the long-term periods doesn't show accurate value compared to the changes in the short-term periods. However, filiration should be conducted carefully. Important information may be lost as a result of the smoothing filter. Therefore, filtration should be carried out in several ways, the results should be verified with the help of correlation analysis.

The second stage is a conduction of the forecast index. For this reason regression model selection and installation is carried out.

Regression analysis is used for two reasons:

1. Detection of relationship between the measured parameters;

2. Prognose of the value of a variable based on the value of regression equation.

Monitoring with carbon monoxide poisoning shows interesting facts according to the method of time series in the monitoring of indicators to determine whether certain moments of time, but also forecast the change indicators. Time series method is using to show the changed indicators of regression equation by time to time. Single regression equation shows the variation of the moments and observation of a person poisoned by a factor. Changes of signs in time creates time series of dynamic rows. The characteristics of that rows is time factor (x), and dependant variable (y) factor, the sign of the value changes. The dependence between them can be shown as a regression equation.

The changes indications by using the method of time series depend on single factor regression equation or multivariate factor of regression equation. In addition, the figure forecast in a single-factor regression equation is given by:

$$y = a + b * x \tag{6}$$

where a is the free member; b determines the slope of the regression line of rectangular axes. According to the least squares method, to determine the parameters, the equations will be as follows:

$$a * n + b \sum x = \sum y \tag{7}$$

$$a \sum x + b \sum x^2 = \sum y * x \tag{8}$$

Formulas given for determination of parameters:

$$a = y - b * x \tag{9}$$

$$b = \frac{y * x - \overline{y} * \overline{x}}{x^2 - \overline{x}^2} \tag{10}$$

Multivariate regression equation is used to monitor and predict the dynamics of change of many traits at the same time:

$$\widehat{y}_x = a + b_1 x_1 + b_2 x_2 + ... + b_m x_m \tag{11}$$

Based on the assumption multivariate regression testing is not possible, dependents become more obvious on the basis of the probabilities. Because the regression coefficients for the various tendencies traits values cause a shift in the regression line, and can change direction. Even one trait value in the presence causes a change in the outcome. Despite it is necessary to monitor the observation of a large number of indicators in carbon monoxide poisoning, more realistic indication of each individual was considered more appropriate to the forecast by the factorial regression. This has been confirmed in numerous experiments. The prognosis by regression equation is given for a certain time after the end of the monitoring period.

In the third stage, the quality of the model should be estimated.

The regression is carried out by adequacy of the model determination:

$$R^2 = \sum_{i=1}^{N} (\widehat{y}_i - \bar{y})^2 / \sum_{i=1}^{N} (y_i - \bar{y})^2. \tag{12}$$

where $\overline{y}_i$ is relevant to x_i the theoretical or estimated value y_i.

Determination coefficient shows variables depending on the degree of compared dispersion. The adequacy of the regression equation is increasing in respect to R^2 high value. Determination coefficient regression model is useful for prediction. The regression equation for the determination of a criteria of Fisher is used:

$$F = \frac{R^2}{1' - R^2} * \frac{n - m - 1}{m} \tag{13}$$

where R is determination coefficient, n is the number of observations, m is the number of parameters in x variables (the number of factors in linear regression model).

This criterion assesses the significance factors included in the regression equation. Calculated F-value of the significance level αup, are compared with 1 and $n - m - 1$ in table value. If the calculated F value exceeds the value of the table, $F \geq F_{table}$, then x factor included in the model is of statistical significance. If the calculated F is less than table value, x variable doesn't affect to y variables changes and the inclusion in the model is inappropriate.

Determination coefficient with the help of correlation is defined as follows:

$$r = \sqrt{R^2} \tag{14}$$

Determination coefficiency, $-1, +1$ varies in correlation coefficient. Determination coefficiency close $+1$ shows close relation of y variables with x factor to prove that indicator is the most significant factor for formalization of consequences. In this regard, the regression model can be used to forecast the indicator.

The indicators selected for monitoring medicine will be:

$$x_i \in X, i = \overline{1, n}$$

where x_i is an indicator.

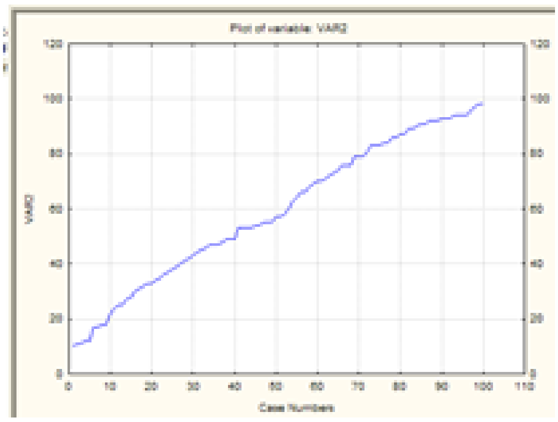
There are ending regulatory values for given parameters. Based on this, there is specific change interval for $\forall x_i$ (in some cases, the standards are different for men and women). Standards in accordance with the upper and lower boundaries is y_i and z_i. Then

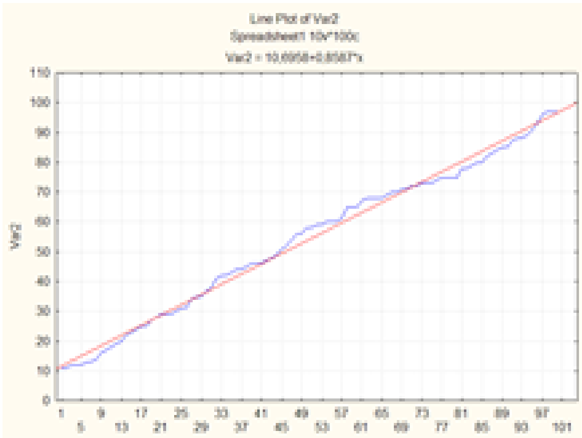
$$y_i < x_i < z_i$$

should be. Each x_i is observed in $T = t_1, t_2, ..., t_k$ time. k is the number of measurements. Then v_i^j can be described as an arbitrary parameter, where $i = 1, 2, ..., n, j = 1, 2, ..., k$. Lower and upper variables can be considered as pathology:

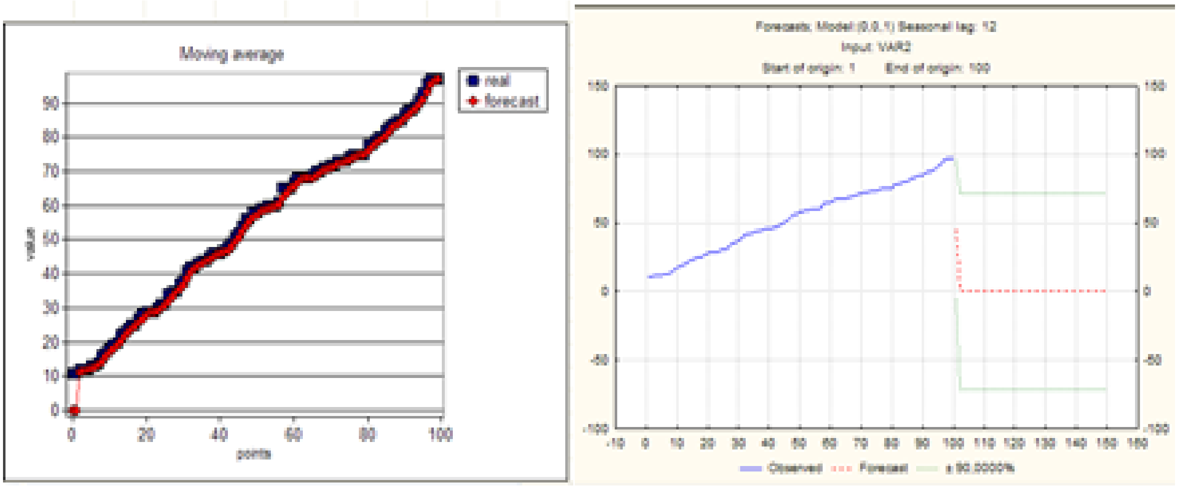
$$x_1^j < y_i \mathrm{x}_1^j > z_i$$

Autocorrelation functions are established for observation of any change of variable x_1^j in $T = t_1, t_2, ..., t_k$ time. It should be noted that the numbers do not reflect the cost of health indicators, the random number generator has been used. K as the number of points used in the determination during the observation period, sometimes it means the number of years or monthes. For example, 100-point numbers with a given distribution (fig. 7a), trend (fig. 7b), smoothing curve (moving average) (fig. 7c), forecasting (fig. 7d), shown a certain time autocorrelation (fig. 7e) and partial autocorrelation function (fig. 7d). This series show ascending value of numbers.

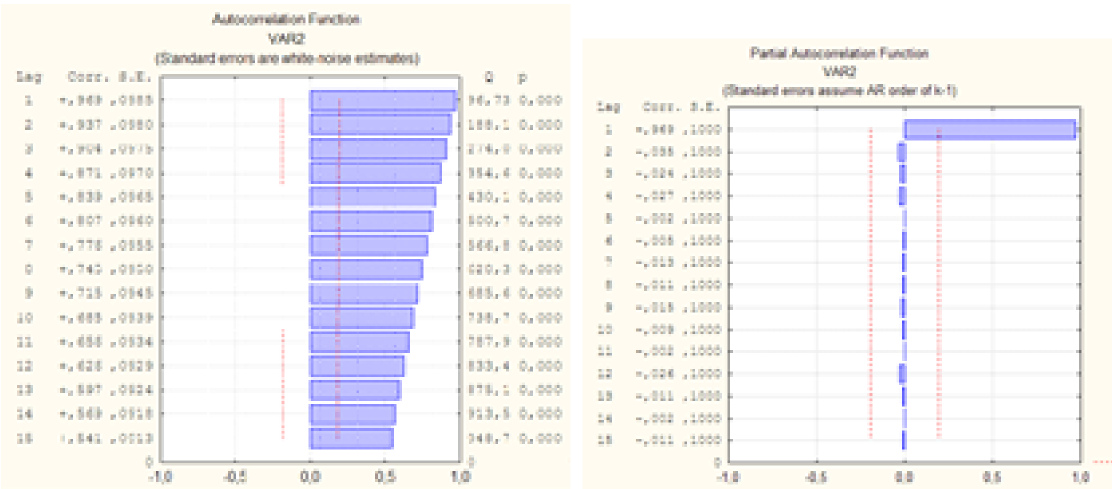




1. b)



1. d)

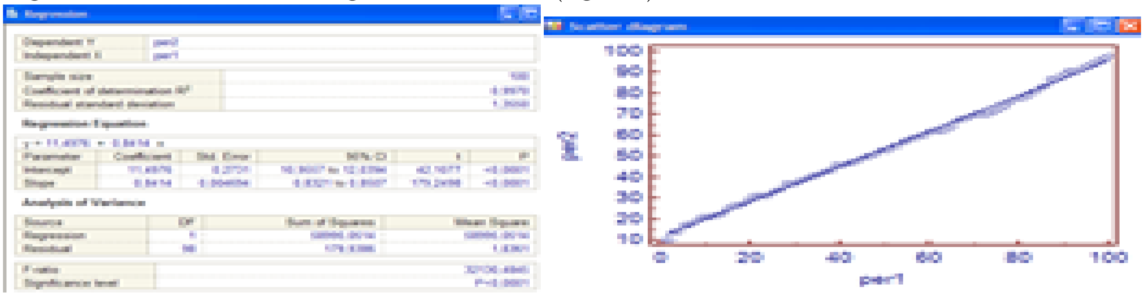


1. f)

Fig.7. Characteristic of time series with ascending numbers

Partial correlation shows variables between two random variables, when taken the effect of internal values of autocorrelation doesn't take into account. Partial autocorrelation is almost same with simple autocorrelation in small moving. In practice, the periodic dependence of the specific autocorrelation is showing as "clean". The appearance of autocorrelation and partial autocorrelation depends on the length of the time series. Autocorrelation function shows the model accurately when the series is long. When the range is short, correlogram loses its accuracy and autocorrelation and autocorrelation estimation degree is decreasing. Meanwhile, the trend shows that there is not a periodic function in autocorrelation changes.

Regression equation for distribution, coefficient of determination (fig. 8a), scatter regression of the order are given as follows (fig. 8b):

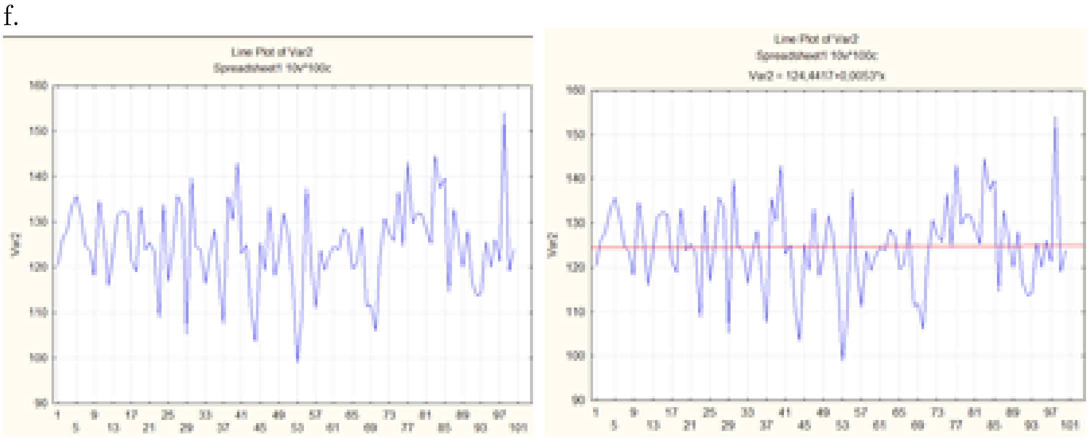


1. b)

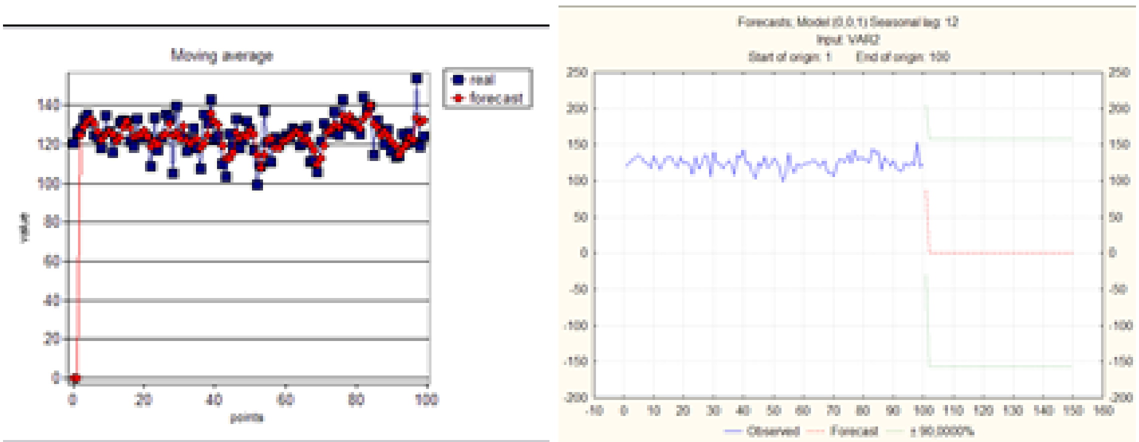
Fig. 8. Regression equation for distribution, coefficient of determination

According to Fisher criteria, this statistics is significant.

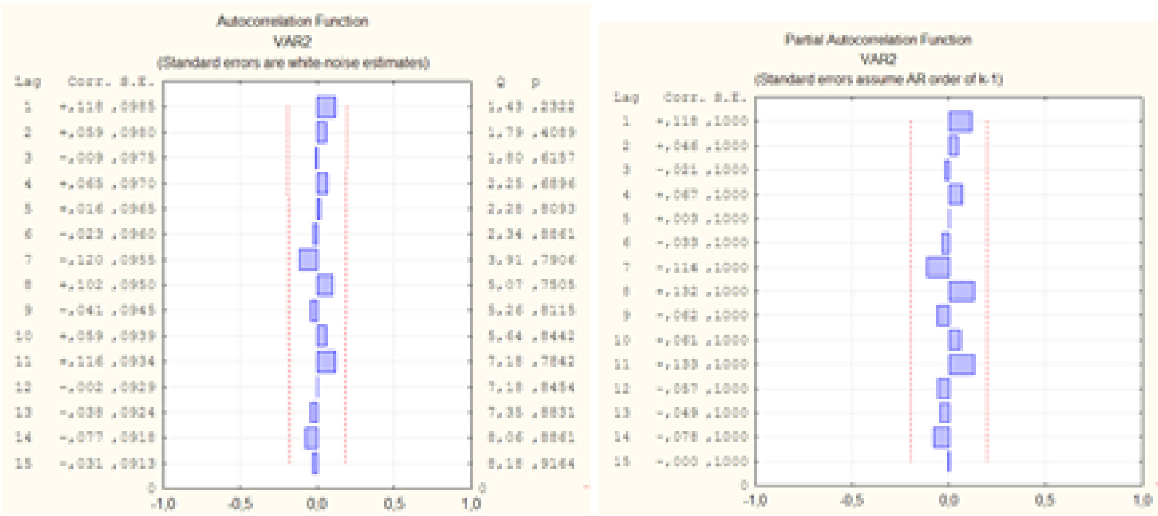
Another example for number of shows with normal distribution in fig. 9 a, b, c, d, e, f.



1. b)



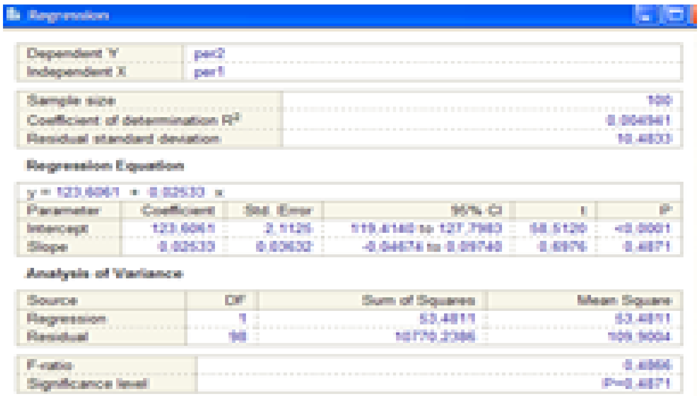
c) d)



e) f)

Fig. 9. Characteristic of time series with normal distribution numbers

Regression equation for distribution, coefficient of determination, scatter and regression of the order are given as follows (fig. 10 a, b):



Regression

Dependent Y	per2
Independent X	per1

Sample size	100
Coefficient of determination R^2	0.004941
Residual standard deviation	10.4833

Regression Equation

y = 123.6061 + 0.02533 x

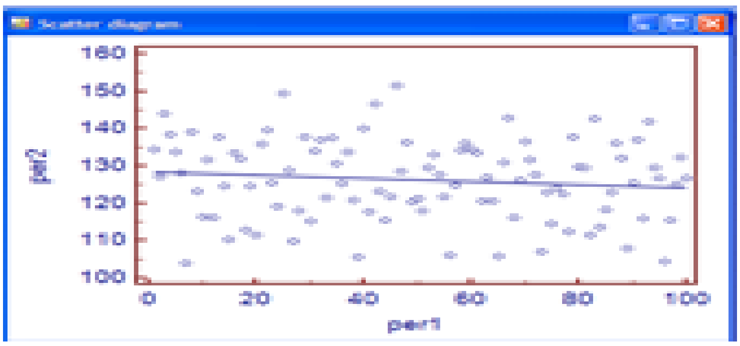
Parameter	Coefficient	Std. Error	95% CI	t	P
Intercept	123.6061	2.1126	119.4140 to 127.7983	58.5120	<0.0001
Slope	0.02533	0.03632	-0.04674 to 0.09740	0.6976	0.4871

Analysis of Variance

Source	DF	Sum of Squares	Mean Square
Regression	1	53.4811	53.4811
Residual	98	10770.2386	109.9004

F-ratio	0.4866
Significance level	P=0.4871

a)



b)

Fig. 10. Regression equation for distribution, coefficient of determination

Smoothing curve, autocorrelation and special autocorrelation functions shows that there is trend in that range. Determination coefficient value shows that forecast is impossibe. According to Fisher theory, the value of indication is not significant.

During the course of the monitoring indicators of each time interval along with the observation of one or several indicators needs to be found observed. Mann-Whitney criteria is used for the evaluation of the difference between two independent indicators, Wilcoxon T-criterion is used for evaluation of monitoring from treatment period, any indication of a change in a certain time, Friedman method is used to measure the difference between double monitoring difference evaluation and Kruskal-Wallis criterion is applied for assessment of presence of indicators in several measurements.

5. Conclusion

In different scientific studies we can see people with carbon monoxide poisoning later suffer from cardiovascular and nervous system diseases. For this reason, patients should be under medical supervision after receiving appropriate treatment. This means conducting a large number of analyses and examinations from time to time.

This work proposes a time series method for monitoring the state of a patient after treatment of carbon monoxide poisoning. The said method allows to trace dynamics of indices in time intervals and detect a more important index for observation of treatment resistant symptoms and elimination of excessive checks. For comparison of the indices in time intervals, parametric and non-parametric criteria of biostatistics are employed.

References

[1] Ernst A and Zibrak JD. Carbon monoxide poisoning. N Engl J Med 1998; 339: 1603–1608.

[2] Wilbur S, Williams M, Williams R, et al. Toxicological profile for carbon monoxide. Atlanta, GA: Agency for Toxic Substances and Disease Registry (US), Table 2-1, Blood carboxyhaemoglobin (COHb) levels corresponding to adverse health effects of carbon monoxide, https://www.ncbi.nlm.nih.gov/books/NBK153692/ table/T10/ (2012, accessed 31 July 2019).

[3] Gozubuyuk AA, Dag H, Kacar A, et al. Epidemiology, pathophysiology, clinical evaluation, and treatment of carbon monoxide poisoning in child, infant, and fetus. North Clin Istanb 2017; 4: 100–107

[4] C Mattiuzzi and G Lippi. Worldwide epidemiology of carbon monoxide poisoning. Human and Experimental Toxicology 2020, Vol. 39(4) 387–392. DOI: 10.1177/0960327119891214

[5] Ren Xiaofan, T. W. *et al.* Risk assessment of public health emergencies concerned in the mainland of China, January 2022. *Dis. Surveill.* 37, 7–11 (2022).

[6] Zhang, L. *et al.* Acute carbon monoxide poisoning in Shandong, China: An observational study. *Chin. Med. J.(Engl.)* https://doi.org/10.1097/CM9.0000000000001942 (2022).

[7] Mingjun Li, Bing Shan, Xiumiao Peng, Huiyun Chang, Liangliang Cui. An urgent health problem of indoor air pollution: results from a 15-years carbon monoxide poisoning observed study in Jinan City. Scientifc Reports. 2023 Jan 28;13(1):1619. doi: 10.1038/s41598-023-28683-0.

[8] Lee OH. Review on the carbon monoxide poisoning in Korea. Kor J Env Hlth Soc 1978;5:25-39 (Korean).

[9] Kim JH, Lim AY, Cheong HK. Trends of accidental carbon monoxide poisoning in Korea, 1951-2018. Epidemiology and Health. Volume: 42. August, 2020. https://doi.org/10.4178/epih.e2020062 (Korean).

[10] Seong Gyu Kim, Jungmin Woo, Gun Woo Kang. A case report on the acute and late complications associated with carbon monoxide poisoning Acute kidney injury, rhabdomyolysis, and delayed leukoencephalopathy. Medicine (Baltimore). 2019 May. 98(19).

[11] E. Yoshioka, S.J.B. Hanley, Y. Kawanishi, Y. Saijo, Epidemic of charcoal burning suicide in Japan, Br. J. Psychiatry 204 (2014) 274–282, https://doi.org/10.1192/bjp.bp.113.135392.

[12] M. Braubach, A. Algoet, M. Beaton, S. Lauriou, M.-E. Heroux, M. Krzyzanowski. Mortality associated with exposure to carbon monoxide in WHO European Member States. International Journal of Indoor Environment and Health. 2013 Apr.; 23(2):115-25.

[13] Office for National Statistics. Number of Deaths From Carbon Monoxide Poisoning, A. Roca-Barceló, et al. Preventive Medicine 136 (2020) 106104 10 England and Wales, 2015 and 2016. 2017.

[14] Health and Safety Executive. Cross government group on gas safety and carbon monoxide. Annual Report 2017/2018. 2019.

[15] Ghosh R. E., Close R., McCann L. J., Crabbe H., Garwood K., Hansell A. L., et al. Analysis of hospital admissions due to accidental non-fire-related carbon monoxide poisoning in England, between 2001 and 2010. J Public Health (Oxf). 2016; 38:76–83.

[16] Roca-Barcelo A., Crabbe H., Ghosh R., Freni-Sterrantino A., Fletcher T., Leonardi G., Hoge C., Hansell A.L., Piel F.B.. Temporal trends and demographic risk factors for hospital admissions due to carbon monoxide poisoning in England. Preventive Medicine. Elsevier. Vol.136, July 2020.

[17] Cohen I., Garis L., Rajabali F., Pike I.. Carbon Monoxide Poisoning, Hospitalizations and Deaths in Canada. A report by the BC Injury Research and Prevention Unit, for the University of the Fraser Valley: Vancouver, BC. October, 2017.

[18] Weaver LK. Carbon monoxide poisoning. In: Moon RE, editor. Hyperbaric Oxygen Therapy Indications. 14th edition. North Palm Beach, FL: Best Publishing Company, 2019;81–104.

[19] Hampson NB. U.S. mortality due to carbon monoxide poisoning, 1999-2014. Accidental and intentional deaths. Ann Am Thorac Soc. 2016;13: 1768–1774.

[20] CDC (2005) Unit intentional, Non ire-related, carbon monoxide exposures United States, 2001-2003. Morbidity and Mortality Weekly Report (MMWR), CDC, USA 54: 36-39.

[21] Litovitz T, Benson BE, Youniss J, Metz E (2010) Determinants of U.S. poison center utilization. Clin Toxicol 48: 449-457.

[22] Iqbal S., Clower J. H., Hernandez S. A., Damon S. A., Yip F. Y.. (2012) A review of disaster-related carbon monoxide poisoning: surveillance, epidemiology, and opportunities for prevention. Am J Public Health 102: 1957-1963.

[23] Bell J, Bronstein A, Clower JH, Iqbal S, Yip FY, et al. (2011) Carbon Monoxide Exposures United States, 2000-2009. Morbidity and Mortality Weekly Report (MMWR), CDC, USA 60: 1014-1017.

[24] Data from the Centers for Disease Control and Prevention (CDC). USA, published March 27, 2017.

[25] G. Can, U. Sayılı, Ö. A. Sayman, Ö. F. Kuyumcu, D. Yılmaz, E. Esen, E. Yurtseven, E. Erginöz. Mapping of carbon monoxide related death risk in Turkey: a ten-year analysis based on news agency records. BMC Public Health Vol. 19, January 2019

[26] Kao, L.W.; Nanagas, K.A. Toxicity associated with carbon monoxide. Clin. Lab. Med. 2006, 26, 99–125. [CrossRef]

[27] Onvlee-Dekker IM, De Vries AC, Ten Harkel AD: Carbon monoxide poisoning mimicking long-QT induced syncope. Arch Dis Child 2007, 92:244-5.

[28] Chang-Teng Wu, Jing-Long Huang, Shao-Hsuan Hsia. Acute carbon monoxide poisoning with severe cardiopulmonary compromise: a case report. Cases Journal. 2009 Jan 14;2(1):52. doi: 10.1186/1757-1626-2-52.

[29] Henry, C.R.; Satran, D.; Lindgren, B.; Adkinson, C.; Nicholson, C.I.; Henry, T.D. Myocardial injury and long-term mortality following moderate to severe carbon monoxide poisoning. JAMA 2006, 295, 398–402. [CrossRef]

[30] Hsu, P.C.; Lin, T.H.; Su, H.M.; Lee, H.C.; Huang, C.H.; Lai, W.T.; Sheu, S.H. Acute carbon monoxide poisoning resulting in ST elevation myocardial infarction: A rare case report. Kaohsiung J. Med. Sci. 2010, 26, 271–275. [CrossRef]

[31] Dileo, P.A.; Tucciarone, M.; Castro, E.R.; Guerrero, M. Late stent thrombosis secondary to carbon monoxide poisoning. Cardiovasc. Revascularization Med. 2011, 12, 56–58. [CrossRef] [PubMed]

[32] Hocagil, H.; Tanrikulu, C.S.; Ülker, V.; Kaya, U.; Koca, L.; Hocagil, A.C. Asymptomatic myocardial injury in a low level of carbon monoxide poisoning. Eurasian J. Emerg. Med. 2015, 14, 91–93. [CrossRef]

[33] Lee, F.Y.; Chen, W.K.; Lin, C.L.; Kao, C.H. Carbon monoxide poisoning and subsequent cardiovascular disease risk: A nationwide population-based cohort study. Medicine 2015, 94, e624. [CrossRef] [PubMed]

[34] Bahng, Y, Baek, K, Park, J.T., Choi, W.J., Kwak, K. Carbon Monoxide Poisoning and Developing Ischemic Heart Disease: A Nationwide Population-Based Nested Case-Control Study. Toxics *2021,* 9*(10), 239;* https://doi.org/10.3390/toxics9100239

[35] Henry TD, Lesser JR, Satran D. Myocardial fibrosis from severe carbon monoxide poisoning detected by cardiac magnetic resonance imaging. Circulation. 2008;118:792.

[36] George B, Ruiz-Rodriguez E, Campbell CL, Leung SW, Sorrell VL. Acute myocardial injury from carbon monoxide poisoning by cardiac magnetic resonance imaging. Eur Heart J Cardiovasc Imaging. 2014;15:466.

[37] Beppu T, Nishimoto H, Fujiwara S, et al. 1 H-magnetic resonance spectroscopy indicates damage to cerebral white matter in the subacute phase after CO poisoning. J Neurol Neurosurg Psychiatry. 2011;82:869e875.

[38] Thom SR, Taber RL, Mendiguren II , Clark JM, Hardy KR, Fisher AB. Delayed neuropsychologic sequelae after carbon monoxide poisoning: prevention by treatment with hyperbaric oxygen. Ann Emerg Med. 1995;25:474e480.

[39] Rose JJ, Wang L, Xu Q, et al. Carbon monoxide poisoning: pathogenesis, management, and future directions of therapy. Am J Respir Crit Care Med. 2017;195: 596e606

[40] Khot S, Walker M, Lacy JM, Oakes P, Longstreth Jr WT. An unsuccessful trial of immunomodulatory therapy in delayed posthypoxic demyelination. Neurocritical Care. 2007;7:253e256.

[41] Gilmer B, Kilkenny J, Tomaszewski C, Watts JA. Hyperbaric oxygen does not prevent neurologic sequelae after carbon monoxide poisoning. Acad Emerg Med. 2002;9:1e8.

[42] Sun Q, Cai J, Zhou J, et al. Hydrogen-rich saline reduces delayed neurologic sequelae in experimental carbon monoxidetoxicity. Crit Care Med. 2011;39: 765e769

[43] Sung Hwa Kim, Yoonsuk Lee, Soo Kang, Jin Hui Paik, Hyun Kim, Yong Sung Cha. Derivation and Validation of a Score for Predicting Poor Neurocognitive Outcomes in Acute Carbon Monoxide Poisoning. JAMA Netw Open. 2022 May; 5(5): e2210552.

[44] Prockop LD. Carbon monoxide brain toxicity: clinical, magnetic resonance imaging, magnetic resonance spectroscopy, and neuropsychological effects in 9 people. J Neuroimaging. 2005;15:144e149.

[45] Kuroda H, Fujihara K, Mugikura S, Takahashi S, Kushimoto S, Aoki M. Altered white matter metabolism in delayed neurologic sequelae after carbon monoxi poisoning: a proton magnetic resonance spectroscopic study. J Neurol Sci. 2016;360:161e169

[46] Eliseeva I.I., Yuzbashev M.M. The general theory of statistics. A textbook for high schools. Moscow: Finance and Statistics, 2004, 656 p. (in Russian)

[47] Gurieva V.M., Kotov Y.B. Analysis of short segments of time series in health problems // Preprint them. R.A H Keldysh, Moscow, 2005. This work was supported by the Russian Foundation for Basic Research (project ? 04-01-00434) (in Russian)

Irada H. Mirzazadeh
Ministry of Science and Education Republic of Azerbaijan
Institute of Mathematics and Mechanics
E-mail: irada811@gmail.com

Received 01 September 2022
Accepted 07 October 2022

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 10, No 2, 2022, December
ISSN 1560-4055

On Completeness of Eigenfunctions of the Spectral Problem

A.Q. Akhmedov, I.Q.Feyzullayev

Abstract. In this paper is studied the spectral problem for a discontinuous second order differential operator with a spectral parameter in conjugation conditions, that arises by solving the problem on vibrations of a loaded string with the free ends. Asymptotic formulas for the eigenvalues and eigenfunctions of the spectral problem are obtained, a theorem on the completeness and minimality of system of eigenvectors of the linearize operator in spaces $L_p \oplus C$ is proved, as well as a theorem on the completeness and minimality in L_p of the system of eigenfunctions of the spectral problem after excluding an arbitrary eigenfunction from it.

Key Words and Phrases: eigenvalues, eigenfunctions, asymptotic formula, biorthogonal system, completeness, minimality.

2010 Mathematics Subject Classifications: Primary 68N19, 68P10, 68T20

---

1. Introduction

Consider a spectral problem

$$y''(x) + \lambda y(x) = 0 \qquad , x \in \left(0, \frac{1}{3}\right) \bigcup \left(\frac{1}{3}, 1\right), \tag{1}$$

$$\left.\begin{aligned} y'(0) =& y'(1) = 0, \\ y(\tfrac{1}{3} - 0) =& y(\tfrac{1}{3} + 0), \\ y'(\tfrac{1}{3} - 0) - y'(\tfrac{1}{3} + 0) =& \lambda m y(\tfrac{1}{3}), \end{aligned}\right\} \tag{2}$$

which arises by solving the problem on vibrations of a loaded string with the free ends [1-3]. The spectral problem corresponding to the problem of oscillation of a loaded string with intersected ends was studied in [4;5] in the case when the load is fixed in the middle of the string, and in [6-8] in the case when the load is fixed at the point $x = \dfrac{1}{3}$. Similar questions for spectral problems corresponding to the case when the load is fixed at one or both ends of the string, were studied by other methods in[9-12].

2. The asymptotic of eigenvalues and eigenfunctions

Let $\lambda=\rho^2$ and introduce the following notation for boundary forms (2).

$$U_\nu(y) = U_{\nu 1}(y) + U_{\nu 2}(y), \nu=\overline{1,4}, \tag{3}$$

where

$$U_{11}(y) = y'(0), U_{12}(y) \equiv 0,$$

$$U_{21}(y) \equiv 0, U_{22}(y) = y'(1),$$

$$U_{31}(y) = y(\frac{1}{3}-0), U_{32}(y) = -\,y(\frac{1}{3}+0),$$

$$U_{41}(y) = y'(\frac{1}{3}-0), U_{42}(y) = -\,y'(\frac{1}{3}+0) - \rho^2 m y(\frac{1}{3}+0).$$

Firstly, let us prove the following theorem.

Theorem 1. *Spectral problem* (1), (2) *has two series of simple eigenvalues:* $\lambda_{1,n} = (\rho_{1,n})^2$ *,* $n = 1, 2, ...,$*è* $\lambda_{2,n} = (\rho_{2,n})^2$ *,* $n = 0, 1, 2, ...,$ *where*

$$\left.\begin{array}{l} \rho_{1,n} = 3\pi n + \frac{3\pi}{2} + O(\frac{1}{n}) \\ \rho_{2,n} = \frac{3\pi n}{2} + \frac{3\pi}{2} + O(\frac{1}{n}). \end{array}\right\} \tag{4}$$

The eigenfunctions $u(x)$, *n=0,1,2,...,* prescribed by formula

$$u_{i,n}(x) = \begin{cases} cos\frac{2\rho_{i,n}}{3}\ cos\rho_{i,n}\ x, x \in \left[0, \frac{1}{3}\right], \\ cos\frac{\rho_{i,n}}{3}\ cos\rho_{i,n}\ (1-x), x \in \left[\frac{1}{3}, 1\right], \end{cases} \quad i = 1, 2; n \in Z^+ \tag{5}$$

correspond to them.

Proof. Take $y_{11}(x) = \sin\rho x, y_{12}(x) = \cos\rho x$ for $x \in \left[0, \frac{1}{3}\right]$ and $y_{21}(x) = \sin\rho(x - \frac{1}{3}), y_{22}(x) = \cos\rho(x - \frac{1}{3})$ for $x \in \left[\frac{1}{3}, 1\right]$ as linear - independent solutions of the equation (1). Eigenfunctions of the problem (1), (2) will be sought in the form

$$y(x,\rho) = \begin{cases} c_{11}y_{11}(x) + c_{12}y_{12}(x), x \in \left[0, \frac{1}{3}\right], \\ c_{21}y_{21}(x) + c_{22}y_{22}(x), x \in \left[\frac{1}{3}, 1\right], \end{cases} =$$

$$= \begin{cases} c_{11}\sin\rho x + c_{12}\cos\rho x, x \in \left[0, \frac{1}{3}\right], \\ c_{21}\sin\rho(x-\frac{1}{3}) + c_{22}\cos\rho(x-\frac{1}{3}), x \in \left[\frac{1}{3}, 1\right]. \end{cases} \tag{6}$$

Let us require that the function $y(x,\rho)$ satisfies the boundary conditions (2) . Then for definition of numbers $c_{j.k}$ we obtain the system of the linear homogeneous equations

$$\begin{cases} C_{11}U_{\nu 1}(y_{11}) + C_{12}U_{\nu 1}(y_{12}) + C_{21}U_{\nu 2}(y_{21}) + C_{22}U_{\nu 2}(y_{22}) = 0, \\ \nu=\overline{1,4}, (or\ \ briefly\ \ \sum_{j,k=1}^{2} C_{kj}U_{\nu j}(y_{jk}) = 0)\ , \end{cases} \tag{7}$$

whose determinant is

$$\Delta(\rho) = \det \|U_{\nu j}(y_{jk})\|\,, j,k= 1,2,\nu=\overline{1,4}$$

Taking into consideration (3), for values of forms $U_{\nu j}(y_{jk})$ we have

$$\begin{cases} U_{11}(y_{11}) =\rho, U_{11}(y_{12}) = 0, U_{12}(y_{21}) = 0, U_{12}(y_{22}) = 0, \\ U_{21}\,(y_{11}) = 0, U_{21}\,(y_{12}) = 0, U_{22}\,(y_{21}) =\rho\cos\frac{2\rho}{3}, U_{22}\,(y_{22}) = -\rho\sin\frac{2\rho}{3}, \\ U_{31}(y_{11}) = \sin\frac{\rho}{3}, U_{31}(y_{12}) = \cos\frac{\rho}{3}, U_{32}(y_{21}) = 0, U_{32}(y_{22}) = -\,1, \\ U_{41}(y_{11}) =\rho\cos\frac{\rho}{3}, U_{41}(y_{12}) = -\,\rho\sin\frac{\rho}{3}, U_{42}(y_{21}) = -\,\rho, U_{42}(y_{22}) = -\,\rho^2 m \end{cases} \tag{8}$$

Opening determinant $\Delta(\rho)$, with the account (8) we obtain

$$\Delta\,(\rho) =\rho^3 \left(\sin\rho \ + \rho m \cos\frac{\rho}{3}\cos\frac{2\rho}{3}\right) \quad . \tag{9}$$

From the formula (9) it is obvious, that function $\Delta\,(\rho)$ has two series of zeros $\rho_{1,n}$ and $\rho_{2,n}$, which are asymptotically close to the zeros of the functions are $\cos\frac{\rho}{3}$ and $\cos\frac{2\rho}{3}$, respectively.

Reasoning as in [13, p.20], we obtain that for $\rho_{1,n}$ and $\rho_{2,n}$ the following asymptotic formula are true

$$\rho_{1,n}= 3\pi n+\frac{3\pi}{2} + O(\frac{1}{n}), \rho_{2,n}=\frac{3\pi n}{2} + \frac{3\pi}{2} + O(\frac{1}{n})$$

Taking into account (8), from (7) we obtain

$$c_{11}= 0 \ ,$$

$$c_{21}\cos\frac{2\rho}{3} - c_{22}\sin\frac{2\rho}{3} = 0 \ ,$$

$$c_{22}=c_{12}\cos\frac{\rho}{3} \ .$$

It is clear that in these relation $\cos\frac{2\rho}{3} \neq 0$ and choosing $c_{12}= \cos\frac{2\rho}{3}$, we get $c_{22}= \cos\frac{2\rho}{3}\cos\frac{\rho}{3}$, $c_{21}= \cos\frac{\rho}{3}\sin\frac{2\rho}{3}$. Taking into account the obtained values of the coefficients in (6), and substituting $\rho = \rho_{1,n}$ and $\rho = \rho_{2,n}$ there, respectively, we get that the eigenfunction $u_{i,n}\,(x)\,,\quad i = 1,2; n \in Z^+$, corresponding to eigenvalue $\lambda_{i,n}= (\rho_{i,n})^2 \ i = 1,2; n \in Z^+$ in the following form

$$u_{i,n}\,(x) = \begin{cases} cos\frac{2\rho_{i,n}}{3}\ cos\rho_{i,n}\ x, x \in \left[0, \frac{1}{3}\right], \\ cos\frac{\rho_{i,n}}{3}\ cos\rho_{i,n}\ (1-x)\,, x \in \left[\frac{1}{3}, 1\right], \end{cases} \quad i = 1,2; n \in Z^+$$

Theorem is proved.

3. Construction of the Green's function and the resolvents of the linearized operator

Now let's pass to construction of the Green's function of problem (1) , (2). It is defined as a kernel of integral representation for solution of the corresponding non-homogeneous problem

$$y''(x) + \rho^2 y(x) = f(x), \tag{10}$$

satisfying boundary conditions (2). The solution of problem (10) ,(2) will be sought in the form

$$y(x) = \begin{cases} y_1(x),\ for\ \ x \in \left[0, \frac{1}{3}\right] \ , \\ y_2(x), for\ x \in \left[\frac{1}{3}, 1\right] \ , \end{cases} \tag{11}$$

where

$$\begin{cases} y_1(x) = c_{11}y_{11}(x) + c_{12}y_{12}(x) + \int_0^{\frac{1}{3}} g(x,\xi,\rho)f(\xi)d\xi, x \in \left[0, \frac{1}{3}\right], \\ y_2(x) = c_{21}y_{21}(x) + c_{22}y_{22}(x) + \int_{\frac{1}{3}}^{1} g(x,\xi,\rho)f(\xi)d\xi, x \in \left[\frac{1}{3}, 1\right]. \end{cases} \tag{12}$$

$$y_{11}(x) = e^{i\rho x}, y_{12}(x) = e^{-i\rho x}\ ,\ x \in \left[0, \frac{1}{3}\right];$$

$$y_{21}(x) = e^{i\rho\left(x-\frac{1}{3}\right)},\ y_{22}(x) = e^{-i\rho\left(x-\frac{1}{3}\right)},\ x \in \left[\frac{1}{3}, 1\right]$$

$$g_1(x,\xi,\rho) = \begin{cases} -\frac{1}{2i\rho}\left(e^{i\rho(x-\xi)} - e^{-i\rho(x-\xi)}\right), 0 \le x < \xi \le \frac{1}{3}, \\ \\ \frac{1}{2i\rho}(e^{i\rho(x-\xi)} - e^{-i\rho(x-\xi)}), 0 \le < x \le \frac{1}{3}, \end{cases} \tag{13}$$

$$g_2(x,\xi,\rho) = \begin{cases} -\frac{1}{2i\rho}(e^{i\rho(x-\xi)} - e^{-i\rho(x-\xi)}), \frac{1}{3} \le x < \xi \le 1, \\ \\ \frac{1}{2i\rho}(e^{i\rho(x-\xi)} - e^{-i\rho(x-\xi)}), \frac{1}{3} \le \xi < x \le 1. \end{cases} \tag{13'}$$

Let us require that the function (11) satisfies the boundary conditions (2). Then for definition of numbers $C_{j,k}$ we obtain the system of algebraic equations

$$U_\nu(y) = \sum_{j,k=1}^{2} C_{j,k}U_{\nu j}(y_{jk}) + \int_0^{\frac{1}{3}} U_{\nu 1}(g)f(\xi)d\xi + \int_{\frac{1}{3}}^{1} U_{\nu 2}(g)f(\xi)d\xi = 0, \nu = \overline{1,4}. \tag{14}$$

Let's define numbers $C_{j,k}$ from (14) and substituting their values in (12), for solving the problem (10), (12) we obtain the formula

$$\begin{matrix} y_1(x) = \int_0^{\frac{1}{3}} G_{11}(x,\xi,\rho)f(\xi)d\xi + \int_{\frac{1}{3}}^{1} G_{12}(x,\xi,\rho)f(\xi)d\xi, x \in \left[0, \frac{1}{3}\right], \\ y_2(x) = \int_0^{\frac{1}{3}} G_{21}(x,\xi,\rho)f(\xi)d\xi + \int_{\frac{1}{3}}^{1} G_{22}(x,\xi,\rho)f(\xi)d\xi, x \in \left[\frac{1}{3}, 1\right], \end{matrix} \tag{14'}$$

where

$$G_{11}(x,\xi,\rho)=\frac{1}{\Delta(\rho)}\begin{vmatrix} g & y_{11} & y_{12} & 0 & 0 \\ U_{\nu1}(g) & U_{\nu1}(y_{11}) & U_{\nu1}(y_{12}) & U_{\nu2}(y_{21}) & U_{\nu2}(y_{22}) \\ \dots & \dots & \dots & \dots & \dots \\ \nu=\overline{1,4} & & & & \end{vmatrix}$$

$$G_{12}(x,\xi,\rho)=\frac{1}{\Delta(\rho)}\begin{vmatrix} 0 & y_{11} & y_{12} & 0 & 0 \\ U_{\nu2}(g) & U_{\nu1}(y_{11}) & U_{\nu1}(y_{12}) & U_{\nu2}(y_{21}) & U_{\nu2}(y_{22}) \\ \dots & \dots & \dots & \dots & \dots \\ \nu=\overline{1,4} & & & & \end{vmatrix}$$

$$G_{21}(x,\xi,\rho)=\frac{1}{\Delta(\rho)}\begin{vmatrix} 0 & 0 & 0 & y_{21} & y_{22} \\ U_{\nu1}(g) & U_{\nu1}(y_{11}) & U_{\nu1}(y_{12}) & U_{\nu2}(y_{21}) & U_{\nu2}(y_{22}) \\ \dots & \dots & \dots & \dots & \dots \\ \nu=\overline{1,4} & & & & \end{vmatrix}$$

$$G_{22}(x,\xi,\rho)=\frac{1}{\Delta(\rho)}\begin{vmatrix} g & 0 & 0 & y_{21} & y_{22} \\ U_{\nu2}(g) & U_{\nu1}(y_{11}) & U_{\nu1}(y_{12}) & U_{\nu2}(y_{21}) & U_{\nu2}(y_{22}) \\ \dots & \dots & \dots & \dots & \dots \\ \nu=\overline{1,4} & & & & \end{vmatrix}$$

and $\Delta(\rho)$ is a determinant from (7).

$$U_{11}\,(g)=-\frac{i\rho}{2i\rho}\left(e^{i\rho\xi}+e^{-i\rho\xi}\right),U_{12}(g)=0$$

$$U_{21}(g)=0,U_{22}(g)=\frac{i\rho}{2i\rho}\left(e^{i\rho(1-\xi)}+e^{-i\rho(1-\xi)}\right)$$

$$U_{31}\,(g)=\frac{1}{2i\rho}\left(e^{i\rho\left(\frac{1}{3}-\xi\right)}-e^{-i\rho\left(\frac{1}{3}-\xi\right)}\right),U_{32}\,(g)=\frac{1}{2i\rho}\left(e^{i\rho\left(\frac{1}{3}-\xi\right)}-e^{-i\rho\left(\frac{1}{3}-\xi\right)}\right) \tag{15}$$

$$U_{41}(g)=\frac{i\rho}{2i\rho}\left(e^{i\rho(\frac{1}{3}-\xi)}+e^{-i\rho(\frac{1}{3}-\xi)}\right),U_{42}(g)=\frac{i\rho}{2i\rho}\left(e^{i\rho(\frac{1}{3}-\xi)}+e^{-i\rho(\frac{1}{3}-\xi)}\right)+$$

$$+\frac{\rho^2 m}{2i\rho}\left(e^{i\rho(\frac{1}{3}-\xi)}-e^{-i\rho(\frac{1}{3}-\xi)}\right)$$

$$U_{11}(y_{11})=i\rho,U_{11}(y_{12})=-i\rho,U_{12}(y_{21})=0,U_{12}(y_{22})=0,$$

$$U_{21}(y_{11})=0,U_{21}(y_{12})=0,U_{22}(y_{21})=i\rho e^{\frac{2i\rho}{3}},U_{22}(y_{22})=-i\rho e^{-\frac{2i\rho}{3}},$$

$$U_{31}(y_{11}) = e^{\frac{i\rho}{3}}, U_{31}(y_{12}) = e^{-\frac{i\rho}{3}}, U_{32}(y_{21}) = -1, U_{32}(y_{22}) = -1,$$

$$U_{41}(y_{11}) = i\rho e^{\frac{i\rho}{3}}, U_{41}(y_{12}) = -i\rho e^{-\frac{i\rho}{3}}, U_{42}(y_{21}) = -i\rho - \rho^2 m, U_{42}(y_{22}) = i\rho - \rho^2 m.$$

Let's substitute (15), (8) and (9) n determinants of formula for $G_{kj}(x,\xi,\rho)$. Transforming the received determinants similar to [14, p. 95], and then opening them, we obtain the formula for the Green's function components. We'll formulate it as a lemma.

Lemma 2. *For the Green's function components $G_{kj}(x,\xi,\rho)$ of the problem* (1), (2) *the following expressions are true:*

$$G_{11}(x,\rho,\xi) = \pm\rho^2 e^{i\rho(x-\xi)} - \frac{\rho^2}{\triangle(\rho)}\left(2e^{-i\rho} - i\rho m e^{-i\rho} = i\rho m e^{-\frac{i\rho}{3}}\right)e^{i\rho x}\cdot e^{i\rho\xi} -$$
$$-\frac{\rho^2}{\triangle(\rho)}\left(2e^{i\rho} + i\rho m e^{-\frac{i\rho}{3}} + i\rho m e^{i\rho}\right)e^{i\rho x}\cdot e^{-i\rho\xi} - \tag{16}$$
$$-\frac{\rho^2}{\triangle(\rho)}\left(2e^{i\rho} + 2i\rho m e^{-\frac{i\rho}{3}} + i\rho m e^{i\rho}\right)e^{-i\rho x}\cdot e^{-i\rho\xi} -$$
$$-\frac{\rho^2}{\triangle(\rho)}\left(2e^{i\rho} + i\rho m e^{i\rho}\right)e^{-i\rho x}\cdot e^{-i\rho\xi}, x\in\left[0,\frac{1}{3}\right], \xi\in\left[0,\frac{1}{3}\right],$$

$$G_{12}(x,\rho,\xi) = 2e^{-\frac{2i\rho}{3}}e^{-i\rho x}e^{-i\rho(\xi-1)}\left(e^{\frac{2i\rho}{3}} - e^{i\rho(\xi-\frac{1}{3})}e^{i\rho(\xi-1)}\right)\left(e^{2i\rho x}+1\right) \tag{17}$$

$$G_{21}(x,\rho,\xi) = 2e^{-i\rho}e^{-i\rho(\xi-\frac{1}{3})}e^{-i\rho(x-\frac{1}{3})}\left(e^{\frac{i\rho}{3}}e^{i\rho(\xi-\frac{1}{3})}e^{i\rho x}+1\right)\left(e^{\frac{4i\rho}{3}} + e^{2i\rho(x-\frac{1}{3})}\right) \tag{18}$$

$$G_{22}(x,\rho,\xi) = \pm\rho^2 e^{i\rho(x-\xi)} + \frac{\rho^2}{\triangle(\rho)}\left((1-i\rho m)e^{\frac{2i\rho}{3}} - (1+i\rho m)e^{\frac{4i\rho}{3}} + e^{\frac{-2i\rho}{3}}\right)e^{i\rho\left(x-\frac{1}{3}\right)}e^{-i\rho\xi} +$$
$$+\frac{\rho^2}{\triangle(\rho)}\left(e^{\frac{-4i\rho}{3}} - (1+i\rho m)e^{\frac{-2i\rho}{3}} - e^{\frac{4i\rho}{3}}\right)e^{i\rho\left(x-\frac{1}{3}\right)}e^{i\rho\xi} - \tag{19}$$
$$-\frac{\rho^2}{\triangle(\rho)}\left(-(1+i\rho m)e^{\frac{2i\rho}{3}} - (-3+i\rho m)e^{\frac{4i\rho}{3}}\right)e^{-i\rho\left(x-\frac{1}{3}\right)}e^{-i\rho\xi} -$$
$$-\frac{\rho^2}{\triangle(\rho)}\left((1+i\rho m) - (1+i\rho m)e^{\frac{2i\rho}{3}}\right)e^{-i\rho\left(x-\frac{1}{3}\right)}e^{i\rho\xi},\ x\in\left[0,\frac{1}{3}\right], \rho\in\left[0,\frac{1}{3}\right].$$

Let us now proceed to the construction of linearizing operator. By $W_p^k\left(0,\frac{1}{3}\right)\oplus W_p^k\left(\frac{1}{3},1\right)$ we denote a space functions whose contractions on segments $\left[0,\frac{1}{3}\right]$ and $\left[\frac{1}{3},1\right]$ belong correspondingly to Sobolev spaces $W_p^k\left(0,\frac{1}{3}\right)$ and $W_p^k\left(\frac{1}{3},1\right)$. Let's define the operator $\mathcal{L}$ in $L_p(0,1)\oplus C$ as follows:

$$D(\mathcal{L}) = \left\{\begin{array}{l} \widehat{u}\in L_p(0,1)\oplus C{:}\widehat{u}=\left(u, mu\left(\frac{1}{3}\right)\right), u\in W_p^2\left(0,\frac{1}{3}\right)\bigcup\left(\frac{1}{3},1\right), \\ u'(0)=u'(1)=0, u\left(-\frac{1}{3}\right)=u\left(\frac{1}{3}\right), \end{array}\right\} \tag{20}$$

and for $\widehat{u} \in D(\mathcal{L})$

$$\mathcal{L}\widehat{u} = \left(-u''; u'\left(\frac{1}{3}-0\right) - u'\left(\frac{1}{3}+0\right), \widehat{u} \in D(\mathcal{L}) \right). \tag{21}$$

Lemma 3. *Operator defined by the formula* (20), (21) *is a linear closed operator with dense definitional domain in* $L_p(0,1) \oplus C$. *Eigenvalues of the operator* $\mathcal{L}$ *and problem* (1), (2) *coincide, and* $\{\widehat{u}_k\}_{k\in N_0}$ *are eigenvectors of the operator* $\mathcal{L}$ *, where* $N_0 = N \bigcup \{0\}$,

$$\widehat{u}_{i,n} = \left(u_{i,n}(x) ; m cos \frac{2\rho_{i,n}}{3} cos \frac{\rho_{i,n}}{3} \right), \; i = 1,2; n \in Z^{+}.$$

Proof. To prove the first part of the lemma we take $(u, \alpha) \in L_p(0,1) \oplus C$ and we define the functional $F(\widehat{u})$ as follows:

$$F(\widehat{u}) = mu\left(\frac{1}{3}\right) - \alpha.$$

Let us assume

$$U_\nu(\widehat{u}) = U_\nu(u), \nu = 1,2,3.$$

Then $F, U_\nu, \nu = 1,2,3$ are bounded linear functionals on $W_p^2\left(0, \frac{1}{3}\right) \bigcup \left(\frac{1}{3}, 1\right) \oplus C$, but unbounded on $L_p(0,1) \oplus C$. Therefore (see, for example, [12, pp. 27-29]) the set

$$D(\mathcal{L}) = \left\{ \widehat{u} = (u, \alpha), u \in W_p^2\left(0, \frac{1}{3}\right) \bigcup \left(\frac{1}{3}, 1\right), F(\widehat{u}) = U_\nu(\widehat{u}) = 0, \nu = 1,2,3 \right\}.$$

is everywhere dense in $L_p(0,1) \oplus C$, and $\mathcal{L}$ is a closed operator as contraction of corresponding closed maximal operator.
The second part of the lemma is verified directly.
The lemma is proved.

For construction resolvent of operator $\mathcal{L}$, consider the equation

$$\mathcal{L}\,\widehat{u} - \lambda \widehat{u} = \overset{\widehat{u}}{f}, \tag{22}$$

where $\widehat{u} \in D(\mathcal{L}), f^{\widehat{u}} = (f, \beta) \in L_p(0,1) \oplus C$. We can rewrite equation (22) in the form of

$$\begin{cases} -u'' = \lambda u + f, \\ u'\left(-\frac{1}{3}\right) - u'\left(\frac{1}{3}\right) - \lambda m u\left(\frac{1}{3}\right) = \beta, \\ U_\nu(u) = 0, \nu = 1,2,3. \end{cases} \tag{23}$$

Lemma 4. *For solution* $\widehat{u} = \left(u, mu\left(\frac{1}{3}\right)\right)$ *of the equation* (22) *it holds the following representations*

$$u(x,\rho) = \frac{\beta\rho^2}{\triangle(\rho)} \left(e^{\frac{2i\rho}{3}} + e^{\frac{-2i\rho}{3}} \right) \left(e^{i\rho x} + e^{-i\rho x} \right) +$$

$$+\int_0^{\frac{1}{3}} G_{11}(x,\rho,\xi)f(\xi)d\xi+\int_{\frac{1}{3}}^1 G_{12}(x,\xi,\rho)f(\xi)d\xi \tag{24}$$

if $x\in\left[0,\frac{1}{3}\right]$.

$$u\left(x,\rho\right)=\frac{\beta\rho^2}{\triangle\left(\rho\right)}\left(e^{\frac{i\rho}{3}}+e^{\frac{-i\rho}{3}}\right)\left(e^{i\rho(x-1)}+e^{-i\rho(x-1)}\right)+$$

$$+\int_0^{\frac{1}{3}} G_{21}(x,\rho,\xi)f(\xi)d\xi+\int_{\frac{1}{3}}^1 G_{22}(x,\xi,\rho)f(\xi)d\xi \tag{25}$$

if $x\in\left[\frac{1}{3},1\right]$

$$u\left(\frac{1}{3},\rho\right)=\frac{\beta\rho^2}{\triangle\left(\rho\right)}\left(e^{\frac{2i\rho}{3}}+e^{\frac{-2i\rho}{3}}\right)\left(e^{\frac{i\rho}{3}}+e^{\frac{-i\rho}{3}}\right)+$$

$$+\int_0^{\frac{1}{3}} G_{21}(x,\rho,\xi)f(\xi)d\xi+\int_{\frac{1}{3}}^1 G_{22}(x,\rho,\xi)f(\xi)d\xi. \tag{26}$$

Proof. The solution of (23) will be sought in the form

$$u(x,\rho)=\begin{cases} C_{11}y_{11}(x)+C_{12}y_{12}(x)+y_1(x), & for \quad x\in\left[0,\frac{1}{3}\right], \\ C_{21}y_{21}(x)+C_{22}y_{22}(x)+y_2(x), & for \quad x\in\left[\frac{1}{3},1\right], \end{cases} \tag{27}$$

where $y_1(x)$ and $y_2(x)$ are defined by (14). Since $y(x)$, defined by (11) satisfies boundary conditions (2), then

$$U_\nu(y)=0,\nu=\overline{1,4}. \tag{28}$$

Let's demand the function $u(x,\rho)$ satisfy boundary conditions $U_\nu(u)=0,\nu=1,2,3,U_\nu(u)=\beta$. Then taking into account (28) from (27) we obtain

$$\begin{cases} C_{11}U_{\nu1}(y_{11})+C_{12}U_{\nu1}(y_{12})+C_{21}U_{\nu2}(y_{21})+C_{22}U_{\nu2}(y_{22})=0,\nu=1,2,3; \\ C_{11}U_{41}(y_{11})+C_{12}U_{41}(y_{12})+C_{21}U_{42}(y_{21})+C_{22}U_{42}(y_{22})=\beta. \end{cases}$$

Solving this system with respect to unknowns C_{kj}, we get

$$C_{11}=-\frac{\beta}{\Delta(\rho)}U_{11}(y_{12})\left[U_{22}(y_{21})U_{32}(y_{22})-U_{22}(y_{22})U_{32}(y_{21})\right],$$

$$C_{12}=\frac{\beta}{\Delta(\rho)}U_{11}(y_{11})\left[U_{22}(y_{21})U_{32}(y_{22})-U_{22}(y_{22})U_{32}(y_{21})\right],$$

$$C_{21}=\frac{\beta}{\Delta(\rho)}U_{22}(y_{22})\left[U_{11}(y_{11})U_{31}(y_{12})-U_{11}(y_{12})U_{31}(y_{11})\right],$$

$$C_{22}=-\frac{\beta}{\Delta(\rho)}U_{22}(y_{21})\left[U_{11}(y_{11})U_{31}(y_{12})-U_{11}(y_{12})U_{31}(y_{11})\right].$$

Substituting the received values of the coefficients C_{kj} in (27) and taking into account formula (8) ,(16)-(19) , we obtain the validity of formulas (24) and (25) . And formula (26) is obtained from (25) (or from (24)) by substitution $x=\frac{1}{3}$.

Lemma is proved.

4. Completeness of the eigenfunctions in spaces $L_p(0,1) \oplus C$ and $L_p(0,1)$

Theorem 2. *System* $\{\hat{u}_i, n\}_{i=1,2;n\in Z^+}$ *of eigenvectors of the operator* $\mathcal{L}$ *is complete in* $L_p(0,1) \oplus C$, $1 < p < \infty$.

Proof. To prove the completeness of the system of eigenfunctions of the operator $\mathcal{L}$ in $L_p(0,1) \oplus C$ we need to get the estimation of the resolvent of the operator $\mathcal{L}$ at great values of $|\rho|$. We will use the following known inequalities

$$|\sin\rho| \le ce^{|\rho|\sin\varphi}, |\cos\rho| \le ce^{|\rho|\sin\varphi}, \tag{29}$$

where $\rho = re^{i\varphi}, 0 \le \varphi \le \pi$. Besides, outside of circles of the same radius δ with centres in zero of $\sin\rho$ the following estimation is true

$$|\sin\rho| \ge m_\delta e^{r\sin\varphi}. \tag{30}$$

From estimations (29) , (30) and from the formula (9) it follows that at great values of $|\rho|$ outside circles $K_{j,n}(\delta) = \{\rho: |\rho - \rho_{j,n}| < \delta\}$ of radius δ with the centres in zero of $\Delta(\rho)$ the following estimation is true

$$|\Delta(\rho)| \ge M_\delta r e^{r\sin\varphi}. \tag{31}$$

Assume $G(\delta) = C \setminus \bigcup_{j,n} K_{j,n}(\delta)$. From the representations (24), (25) considering the inequalities (29) –(31) we obtain the inequality

$$|u(x,\rho)| \le \frac{C_\delta}{|\rho|}, \rho \in G(\delta) \ , \quad |\rho| \ge r_0,$$

which fairly uniform on $x \in [0,1]$. From the last estimation it follows that for the resolvent $R(\lambda) = (\mathcal{L} - \lambda I)^{-1}$ of the operator $\mathcal{L}$ outside of the above-stated circles the following estimation is true

$$\left\|R(\rho^2)\right\| \le \frac{C_\delta}{|\rho|}, |\rho| \ge r_0. \tag{32}$$

Having estimation (32), by a standard method (see for example [16]) we obtain that eigenfunctions of operator $\mathcal{L}$ form a complete system in $L_p(0,1) \oplus C$.

Let us note that the system $\{\widehat{u}_{i,n}\}_{i=1,2;n\in Z^+}$ of eigenvectors of the operator $\mathcal{L}$ has a biorthogonal-conjugate system $\{\widehat{v}_{i,n}\}_{i=1,2;n\in Z^+}$, that are the system of eigenvectors of the conjugate operator $\mathcal{L}^*$, which in its turn is the linearized operator of the conjugate spectral problem:

$$v'' + \lambda v = 0, x \in \left(0, \frac{1}{3}\right) \bigcup \left(\frac{1}{3}, 1\right), (1^*)$$

$$v'(0) = v'(1) = 0,$$

$$v\left(\frac{1}{3} - 0\right) = v\left(\frac{1}{3} + 0\right), (2^*)$$

$$v'\left(\frac{1}{3}-0\right)-v'\left(\frac{1}{3}+0\right)=\lambda\overline{m}v\left(\frac{1}{3}\right).$$

Taking into account this by Theorem 2 we obtain

Corollary. *System* $\{\hat{u}_i,n\}_{i=1,2;n\in Z^+}$ *of the eigenvectors of the operator* $\mathcal{L}$ *is complete and minimal in* $L_p(0,1)\oplus C$, $1<p<\infty$.

Now let us consider the completeness and minimality of a system $\{u_i,n\}_{i=1,2;n\in Z^+}$ of eigenfunctions of the problem (1), (2). It is clear that this system is overflowing in $L_p(0,1)$: One function of this system is unnecessary. Let us clarify the following question: Which function can be excluded from the system while maintaining the properties of completeness and minimality and which can not? The answer to this question is given by the following theorem.

Theorem 3. *Let* n_0 *be* some number from the set of indexes Z^+ . Then the system obtained from the system $\{\hat{u}_i,n\}_{i=1,2;n\in Z^+}$ by excluding on arbitrary function u_{i,n_0} is complete and minimal in the space $L_p(0,1)$, $1<p<\infty$

Proof. According to the above mentioned, the system $\{u_i,n\}_{i=1,2;n\in Z^+}$ has a biortogonal-conjugated system $\{\hat{v}_i,n\}_{i=1,2;n\in Z^+}$, where

$$\hat{v}_n=\left(v_n(x);\,\overline{m}v_n\left(\frac{1}{3}\right)\right),$$

and $v_n(x)$ are eigenfunctions of the adjoint problem (1^*), (2^*) . Carrying out similar calculations for eigenfunctions, we find that the formulas are valid

$$v_{i,n}(x)=\begin{cases} c_{\mathrm{i},n}cos\frac{2\rho_{i,n}}{3}\ cos\rho_{i,n}\ x, x\in\left[0,\frac{1}{3}\right], \\ c_{\mathrm{i},n}cos\frac{\rho_{i,n}}{3}\ cos\rho_{i,n}\ (1-x), x\in\left[\frac{1}{3},1\right], \end{cases} \quad i=1,2;n\in Z^+$$

where $c_{\mathrm{i},n}-$ are normalization numbers. From this formula it is clear that if n_0 is number, from the set of index Z^+ then $v_{i,n_0}\left(\frac{1}{3}\right)\neq 0$. Therefore, all statements of the theorem follow from the results of [17] (see also [18]).

References

[1] A.N. Tikhonov, A.A. Samarskii, *Equations of mathematical physics*, Courier Corporation, 2013.

[2] F.V. Atkinson, *Discrete and continuous boundary problems*, Academic Press, New York, London, 1964.

[3] L. Collatz, *Eigenwertaufgaben mit technischen Anwendungen*, Leipzig,1963

[4] T.B. Gasymov, Sh.J. Mammadova, *On convergence of spectral expansions for one discontinuous problem with a spectral parameter in the boundary condition*, Trans.of NAS of Azerb., **XXVI**(4), 103–116.

[5] T.B. Gasymov, A.A. Huseynli, *On completeness of eigenfunctions and associated functions of a discontinuous second order differential operator*, Dokl. NAN of Azerb., **LXVIII**(1), 3–7.

[6] N.Yu. Kapustin, E.I. Moiseev, *The basis property in L_p of the systems of eigenfunctions corresponding to two problems with a spectral parameter in the boundary condition*, Differential equations, **36**(10), 2000, 1498–1501.

[7] N.Yu. Kapustin, E.I. Moiseev, *On spectral problems with a spectral parameter in the boundary condition*, Diff. Uravn., **33**(1), 1997, 16–21.

[8] N.B. Kerimov, V.S. Mirzoyev, *On basis properties of one spectral problem with spectral parameter in the boundary condition*, Sib. Math. Jour., 2003,**44**(5), 1041–1045. (Russian)

[9] N.B. Kerimov, R.G. Poladov, *On basicity in $L_p(0,1), (1 < p < \infty)$ of the system of eigenfunctions of one boundary value problem.* II. Proc. of IMM of NAS of Azerb., **XXIII**, 2005, 65–76.

[10] B.M. Levitan, I.S. Sargsjan, *Introduction to spectral theory*, American Mathematical Society, 1975.

[11] M.A. Naimark, *Linear Differential operators*, Frederik Ungar Publ. Co., New York, 1968.

[12] V.E. Lyantse, O.G. Storozh, *Methods of the theory of unbounded operators*, Kiev, 1983.

[13] N. Dunford, J.T. Schwartz, *Linear operators*, Part 3, New York, London, Sydney, Toronto, 1971.

[14] T.B. Gasymov, *On convergence of spectral expansions for one discontinuous problem with a spectral parameter in the boundary condition*, Trans. of NAS of Azerb., **XXVI**(4), 103–116.

[15] T.B. Gasymov, T.Z. Garayev, *On convergence of spectral expansions for one discontinuous problem with spectral parameter in the boundary condition*, Trans. of NAS of Azerb., **XXVI**(4), 103–116.

[16] B.T. Bilalov, T.R. Muradov, *Defective bases of Banach spaces*, Proc. IMM NAS Azerb., **XXII**, 2005, 23–26.

Alirza Q. Akhmedov
Baku State University, Az1148, Baku, Azerbaijan
E-mail: ehmedov-elirza@mail.ru

Ilkin Q.Feyzullayev
Institute of Mathematics and Mechanics of NAS of Azerbaijan, Az1141, Baku, Azerbaijan
E-mail: ilkin0704@mail.ru

Received 05 September 2022
Accepted 22 October 2022

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 10, No 2, 2022, December
ISSN 1560-4055

Basis Property in L_p of Root Functions of Some Fourth-Order Eigenvalue Problem

Gunay T. Mamedova

Abstract. In this paper, we consider the eigenvalue problem for ordinary differential equations of fourth order, two of the boundary conditions depend on the spectral parameter. We find sufficient conditions under which the system of root functions of this problem forms a defect basis in L_p, $1 < p < \infty$, with a defect number 2.

Key Words and Phrases: eigenvalue problem, spectral parameter, eigenvalue, eigenfunction, defect basis
2010 Mathematics Subject Classifications: 34B05, 34B08, 34C10, 34L10, 47A75, 74H45

1. Introduction

We consider the following eigenvalue problem

$$\ell(y) \equiv y^{(4)}(x) - (q(x)y')' = \lambda y(x)), \ x \in (0, l), \tag{1.1}$$

$$y(0) = y'(0) = 0, \tag{1.2}$$

$$y''(1) - (a\lambda + b)y'(1) = 0, \tag{1.3}$$

$$Ty(1) - c\lambda y(1) = 0, \tag{1.4}$$

where $\lambda \in \mathbb{C}$ is a spectral parameter, $Ty \equiv (py'')' - qy'$, q is nonnegative absolutely continuous function on $[0, 1]$, a, b and c are real constants such that $a > 0$, $b < 0$ and $c > 0$.

The location of eigenvalues on the real axis, the structure of the root subspaces, oscillation properties of eigenfunctions of problem (1.1)-(1.4) and their derivatives was investigated in [10]. Recall that this problem describes the flexural vibrations of a homogeneous rod, in the cross sections of which the longitudinal force acts, the left end is fixed rigidly, and the right end resiliently fastened and on this end an inertial mass is concentrated (see [9, 10]). Obviously, to study this problem of mechanics, we need to study the basic properties of the root functions of problem (1.1)-(1.4) in various function spaces.

Note that the spectral properties, including the basis properties of root functions in the Lebesgue spaces of linear eigenvalue problems for ordinary differential equations of fourth order, were considered in [1-5, 11, 12] (see also references therein). In the present paper, we study the basis properties in the space $L_p(0,1)$, $1<p<\infty$, of root functions of the problem (1.1)-(1.4). We show that, under certain conditions, the system of root functions of this problem, after removing two functions, forms a basis in the space $L_p(0,1)$, $1<p<\infty$, i.e. the system of root functions forms a defect basis in $L_p(0,1)$, $1<p<\infty$, with a defect number 2.

2. Preliminary

Let $H=L_2(0,1)\oplus\mathbb{C}^2$ be a Hilbert space with the scalar product

$$(\hat{y},\hat{\vartheta})_H=\int_0^1 y(x)\overline{\vartheta(x)}dx+|a|^{-1}m\bar{s}+|c|^{-1}\,n\bar{t},$$

where

$$\hat{y}=\{y,m,n\}\in H,\ v=\{\vartheta,s,t\}\in H.$$

We define in the space H an operator

$$\hat{L}\hat{y}=\hat{L}\{y,m,n\}=\{\ell(y),y''(1)-by'(1),Ty(1)\}$$

with the domain

$$D(\hat{L})=\big\{y=\{y,m,n\}\in H: y\in W_4^2(0,1),\ell(y)\in L_2(0,1),y(0)=y'(0)=0,\\ m=ay'(1),n=cy(1)\}$$

which is dense in H [12].

Obviously, the operator $L$is well defined. By direct verification, we conclude that problem (1.1)-(1.4) is equivalent to the following spectral problem

$$L\hat{y}=\lambda\hat{y},\ \hat{y}\in D(L). \tag{2.1}$$

This means that the eigenvalues λ_k, $k\in\mathbb{N}$ of problem (1.1)-(1.4) and problem (2.1) coincide, and there is a one-to-one correspondence between the root functions of these problems

$$y_k\leftrightarrow\hat{y}_k=\{y_k,m_k,n_k\},\ m_k=ay_k'(1),\ n_k=cy_k(1),k\in\mathrm{N}.$$

The problem (1.1)-(1.4) is strongly regular in the sense of [12]. Therefore, the spectrum of this operator is discrete.

Note that the operator L is a non-self-adjoint, closed, compact resolvent operator in H [6, 7].

Since the operator L is not self-adjoint in H, let us introduce the operator $J : H \to H$ by

$$J\hat{y} = \{y, m, n\} = \{y, m, -n\}.$$

The operator J is unitary and symmetric in H. The spectrum of this operator consists of two eigenvalues: -1 with multiplicity 1 and 1 with infinite multiplicity [8]. Hence this operator generates a Pontryagin space $\Pi_1 = L_2(0,1) \oplus C^2$ (see [6]) whose inner product is defined as

$$(\hat{y}, \hat{\vartheta})_1 = (\{y, m, n\}, \{\vartheta, s, t\})_1 = (J\hat{y}, \hat{\vartheta})_H = \int_0^1 y(x)\overline{\vartheta(x)}dx + a^{-1}m\bar{s} - c^{-1}n\bar{t}.$$

Suppose that the operator L^* is an adjoint to the operator L in H.

By [10, Theorem 2.1] the operator L is self-adjoint in Π_1. In view of [7, Section 3, Proposition 5^0], we have

$$L^* = JLJ. \tag{2.2}$$

By virtue of [7, § 4, Theorem 4.2] the system of root vectors $\{\hat{y}_k\}_{k=1}^{\infty}$, $\hat{y}_k = \{y_k, m_k, n_k\}$, $m_k = ay_k'(1)$, $n_k = cy_k(1)$, of the operator L forms an unconditional basis in H.

We introduce the following boundary condition

$$y(1)\cos\delta - Ty(1)\sin\delta = 0, \tag{2.3}$$

where $\delta \in \left[0, \frac{\pi}{2}\right]$.

Along with problem (1.1)-(1.4) we shall consider the eigenvalue problem (1.1)-(1.3), (2.3). It is known [1] that the eigenvalues of problem (1.1)-(1.3), (2.3) are real and positive, and form an infinitely increasing sequence $\{\lambda_{k,\delta}\}_{k=1}^{\infty}$. Moreover, (i) the eigenfunction $y_{k,\delta}(x)$ corresponding to the eigenvalue $\lambda_{k,\delta}$ has exactly $k-1$ simple zeros in the interval $(0,1)$ for $a\lambda_{k,\delta} + b \leq 0$, and either $k-2$ or $k-1$ simple zeros for $a\lambda_{k,\delta} + b \leq 0$; (ii) the function $y_{k,\delta}'(x)$ has exactly $k-1$ simple zeros in the interval $(0,1)$.

It follows from the maximal-minimal property of eigenvalues that

$$\lambda_{1,\frac{\pi}{2}} < \lambda_{1,\pi} < \lambda_{2,\frac{\pi}{2}} < \lambda_{2,\pi} < \ldots < \lambda_{k,\frac{\pi}{2}} < \lambda_{k,\pi} < \ldots\ . \tag{2.4}$$

Theorem 2.1 [10, Theorem 3.2]. *For each fixed $\lambda \in \mathbb{C}$ problem (1.1)-(1.3) has a unique non-trivial solution $y(x,\lambda)$ with a constant factor. Moreover, $y(x,\lambda)$ for each fixed $x \in [0,1]$ is entire function of λ.*

Let $B_k = (\lambda_{k-1,\pi}, \lambda_{k,\pi})$, $k \in \mathbb{N}$, where $\lambda_{0,\pi} = -\infty$.

According to above arguments, Theorem 2.1 and relation (2.3), it is clear that the function

$$G(\lambda) = \frac{Ty(1,\lambda)}{y(1,\lambda}$$

is a meromorphic function of finite order defined on the set

$$B = (\mathbb{C}\backslash\mathbb{R}) \cup \left(\bigcup_{k=1}^{\infty} B_k\right).$$

The eigenvalues $\lambda_{k,\frac{\pi}{2}}$ and $\lambda_{k,\pi}$ of problem (1.1)-(1.3) are zeros and poles of this function, respectively.

Lemma 2.1. [10, Lemmas 3.1, 3.2]. *The following relations hold:*

$$\frac{dG(\lambda)}{d\lambda} = -\frac{1}{y^2(1,\lambda)}\left\{\int_0^1 y^2(x,\lambda)dx + ay'^2(1,\lambda)\right\}, \lambda \in B,$$

$$\lim_{\lambda\to-\infty} G(\lambda) = -\infty.$$

Lemma 2.2. [10, Lemma 3.3]. *For the function $G(\lambda)$, the following representation holds:*

$$G(\lambda) = G(0) + \sum_{k=1}^{\infty} \frac{\lambda\varsigma_k}{\lambda_k(\pi)(\lambda - \lambda_k(\pi))},$$

where $\varsigma_k = \underset{\lambda=\lambda_k(\pi)}{res} G(\lambda)$ *and* $\varsigma_k < 0,\ k \in \mathbb{N}$.

Using Theorem 2.1 and Lemmas 2.1-2.3 in the paper [10] for problem (1.1)-(1.4) the following result is obtained.

Theorem 2.2 [10, Theorem 4.1]. *The eigenvalues of problem (1.1)-(1.4) are real and simple, and form an infinite increasing sequence* $\{\lambda_k\}_{k=1}^{\infty}$ *such that*

$$\lambda_1 \in (-\infty, 0) \text{ and } \lambda_k \in (\lambda_{k-1,\frac{\pi}{2}}, \lambda_{k,\pi}) \text{ for } k = 2, 3, \dots . \tag{2.5}$$

For each $k \in \mathbb{N}$ let y_k be the eigenfunction corresponding to the eigenvalue λ_k of problem (1.1)-(1.4).

Theorem 2.3 *For sufficiently large $k \in \mathbb{N}$ the following asymptotic formulas hold:*

$$\sqrt[4]{\lambda_k} = \left(k - \frac{3}{2}\right)\pi + O\left(\frac{1}{k}\right), \tag{2.6}$$

$$\frac{y_k(1)}{y'_k(1)} = \frac{b}{c}\sqrt[4]{\lambda_k}\left(1 + \frac{1}{c\sqrt[4]{\lambda_k}} + O\left(\frac{1}{k^2}\right)\right). \tag{2.7}$$

The proof of this theorem is similar to that of [5, Theorems 4.2 and formula (5.18)].

3. Basis property of eigenfunctions of problem (1.1)-(1.4)

Suppose that the system $\{\hat{\vartheta}_k\}_{k=1}^{\infty}$, $\hat{\vartheta}_k = \{\vartheta_k, s_k, t_k\}$, is an adjoint system to the system $\{\hat{y}_k\}_{k=1}^{\infty}$, $\hat{y}_k = \{y_k, m_k, n_k\}$. Since all eigenvalues of problem (1.1)-(1.4) are real and simple, according to Lemma 2.1 we get

$$\delta_k = (\hat{y}_k, \hat{y}_k)_1 = ||y_k||_2^2 + a{y'_k}^2(1) - cy_k^2(1) \neq 0. \tag{3.1}$$

Then, analogously to [5, Lemma 5.1], we can show that

$$\hat{\vartheta}_k = \delta_k^{-1} J \hat{y}_k, \ k \in \mathrm{N}. \tag{3.2}$$

Now suppose that r and l $(r \neq l)$ are arbitrary natural numbers. By (2.1), (2.2) and (3.1), (3.2) we have the following relation

$$\Delta_{r,l} = \begin{vmatrix} s_r & s_l \\ t_r & t_l \end{vmatrix} = \delta_r^{-1}\delta_l^{-1} \begin{vmatrix} m_r & m_l \\ -n_r & -n_l \end{vmatrix} =$$

$$-\delta_r^{-1}\delta_l^{-1} \begin{vmatrix} ay_r'(1) & ay_l'(1) \\ cy_r(1) & cy_l(1) \end{vmatrix} =$$

$$= -ac\delta_r^{-1}\delta_l^{-1} \begin{vmatrix} y_r'(1) & y_l'(1) \\ y_r(1) & y_l(1) \end{vmatrix} = -acy_r'(1)y_l'(1) \begin{vmatrix} 1 & 1 \\ \frac{y_r(1)}{y_r'(1)} & \frac{y_l(1)}{y_l'(1)} \end{vmatrix} =$$

$$= -acy_r'(1)y_l'(1) \left\{ \frac{y_l(1)}{y_l'(1)} - \frac{y_r(1)}{y_r'(1)} \right\}. \tag{3.3}$$

Remark 3.1. It follows from [3, Theorem 4.1] that the condition $\Delta_{r,l} \neq 0$ is necessary and sufficient in order to the system of eigenfunctions $\{y_k(x)\}_{k=1,\, k\neq r,l}^{\infty}$ of problem (1.1)-(1.4) to form a basis in the space $L_p(0,1)$, $1 < p < \infty$, (an unconditional basis for $p = 2$).

The main result of this paper is the following theorem.

Theorem 3.1. *Suppose that r and l $(r < l)$ are arbitrarily large enough natural numbers. Then the system of eigenfunctions $\{y_k(x)\}_{k=1,\, k\neq r,l}^{\infty}$ of problem (1.1)-(1.4) forms a basis in space $L_p(0,1)$, $1 < p < \infty$, which is an unconditional basis in $L_2(0,1)$.*

Proof. If r and l $(r < l)$ are arbitrarily large enough natural numbers, then by (2.6) and (2.7) from (3.3) we get

$$-\Delta_{r,l} = acy_r'(1)y_l'(1) \left\{ \frac{y_l(1)}{y_l'(1)} - \frac{y_r(1)}{y_r'(1)} \right\} =$$

$$acy_r'(1)y_l'(1)\frac{b}{c} \left\{ \sqrt[4]{\lambda_l}\left(1 + \frac{1}{c\sqrt[4]{\lambda_l}} + O\left(\frac{1}{l^2}\right)\right) - \sqrt[4]{\lambda_r}\left(1 + \frac{1}{c\sqrt[4]{\lambda_r}} + O\left(\frac{1}{r^2}\right)\right) \right\} =$$

$$= -acy_r'(1)y_l'(1)\frac{b}{c} \left\{ \sqrt[4]{\lambda_l} + \frac{1}{c} + O\left(\frac{1}{l}\right) - \sqrt[4]{\lambda_r} - \frac{1}{c} + O\left(\frac{1}{r}\right) \right\} =$$

$$= acy_r'(1)y_l'(1)\frac{b}{c} \left\{ \left(l - \frac{3}{2}\right)\pi + O\left(\frac{1}{l}\right) - \left(r - \frac{3}{2}\right)\pi + O\left(\frac{1}{r}\right) \right\} >$$

$$> acy'_r(1)y'_l(1)\frac{b}{c}\left\{(l-r)\pi - \frac{K}{l} - \frac{K}{r}\right\} > 0,$$

where $K > 0$ is some constant. Now the assertion of this theorem follows from Remark 3.1 by virtue of the last relation. The proof of this theorem is complete.

References

[1] Z.S. Aliyev, Basis properties of a fourth order differential operator with spectral parameter in the boundary condition, *Cent. Eur. J. Math.* **8**(2) (2010), 378-388.

[2] Z.S. Aliev, Basis properties in L_p of systems of root functions of a spectral problem with spectral parameter in a boundary condition, *Differ. Equat.* 47(6) (2011), 766-777.

[3] Z.S. Aliev, A.A. Dunyamalieva, Defect basis property of a system of root functions of a Sturm-Liouville problem with spectral parameter in the boundary conditions, *Differ. Equat.* **51**(10) (2015), 1249-1266.

[4] Z.S. Aliyev, S.B. Guliyeva, Properties of natural frequencies and harmonic bending vibrations of a rod at one end of which is concentrated inertial load, *J. Differential Equations* **263**(9) (2017), 5830-5845.

[5] Z.S. Aliyev, N.B. Kerimov, V.A. Mehrabov, Convergence of eigenfunction expansions for a boundary value problem with spectral parameter in the boundary conditions. I, *Differ. Equat.* 56(2) (2020), 143-157.

[6] T.Ya. Azizov, I.S. Iokhvidov, Linear operators in Hilbert spaces with G-metric, *Russ. Math. Surv.* **2**6(4) (1971), 45-97.

[7] T.Ya. Azizov, I.S. Iokhvidov, Linear operators in spaces with an indefinite metric, John Vielcy, Chichester, UK (1989).

[8] P.A. Binding, P.J. Browne, Application of two parameter eigencurves to Sturm-Liouville problems with eigenparameter-dependent boundary conditions, *Proc. R. Soc. Edinb., Sect. A* **125**(6) (1995). 1205-1218.

[9] B.B. Bolotin, Vibrations in technique: Handbook in 6 volumes, The vibrations of linear systems, I, Engineering Industry, Moscow (1978).

[10] G.T. Mamedova, Some spectral properties of an eigenvalue problem with spectral parameter contained in the boundary conditions, *Transactions of NAS of Azerbaijan, Issue Mathematics, Series of Physical-Technical and Mathematical Sciences* **38**(1), (2018), 114-122.

[11] E.M. Russakovskii, Operator treatment of boundary problems with spectral parameters entering via polynomials in the boundary conditions, *Funct. Anal. Appl.* **9**(4) (1975), 358-359.

[12] A.A. Shkalikov, Boundary problems for ordinary differential equations with parameter in the boundary conditions, *J. Sov. Math.* **33** (1986), 1311-1342.

Gunay T. Mamedova
Ganja State University, Ganja, Azerbaijan
E-mail: gunaymamedova614@gmail.com

Received 10 September 2022
Accepted 01 November 2022